

PKZIP 2

For DOS

PKZIP

PKUNZIP

PKSFX

**User's Guide, Tutorial
& Command Reference**

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U.S. Patent No. 5,051,745

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PKZIP® Quick Command Reference

Option	See Page(s)	Description
-a+	30	Turn off archive attribute after operation
-b<drive:path>	30	Create temporary .ZIP file on alternate drive
-c	31	Create/Edit comments for all files
-C	31	Add comments for new files only
-d	32	Delete files
-e[xx x n f s 0]	32, 23	Specify compression method
-f	33	Freshen files in a .ZIP file
-h	34	Display command summary (Help)
-i	34	Add files that have the archive attribute set
-j<h,r,s,a>	34	Specify masking of file attributes (default)
-J<h,r,s,a>	34	Specify retention of file attributes
-k	35	Retain original .ZIP file date
-m[u f]	35	Move files (with update freshen)
-o	36	Set .ZIP file date to newest file in ZIP
-P	36	Preserve recursed and specified path names
-q	37	Enable ANSI codes in comment
-r	37	Recurse subdirectories
-rp	37	Preserve recursed path names
-s[password]	38, 68	Encrypt files
-t<date>	38	Take files NEWER than or EQUAL to date -t with no options=take files with today's date.
-T<date>	38	Take files OLDER than date
-u	39	Update files
-v[see page]	40	View file information
-vt[see page]	40	View technical information about file
-w<h,s>	42	Take hidden, system files
-W<h,s>	43	Skip hidden, system files (default)
-x<filename>	43	Exclude file
-x@listfile	43	Exclude list file
-z	43	Create or modify .ZIP file comment
-!	44, 78	Add authenticity verifiable data to .ZIP file
-&	45	Create multi-volume (spanned) archive
-\$[d]	46	Store volume label of drive [d] as part of .ZIP file
-@listfile	46	Generate list file
-=	47	Open files in compatibility mode

PKUNZIP® Quick Command Reference

Option	See Page(s)	Description
-c[m]	48	Extract files to console [with more]
-d	48	Restore stored directory structure and/or create directory specified on command line
-e[r][c d e n p s]	49	Extract files in [reverse] sorted order
-f	50	Freshen files in extract directory
-h	50	Display command summary (Help)
-j<h,r,s,a>	51	Specifies masking of file attributes (default)
-J<h,r,s,a>	51	Specifies extraction of file attributes
-n	51	Extract only newer files
-o	51	Overwrite previously existing files
-p[a / b][c][1-9]	52	Extract files to printer
-q	52	Enable ANSI comments
-s[password]	53, 68	Decrypt files
-t	54	Test .ZIP file integrity
-v[see page]	55, 40	View file information
-vt[see page]	55, 40	View technical information about file
-x<filespec>	55	Exclude file from extraction
-x@listfile	55	Exclude list files from extraction
-\$	55	Restore volume label
-@listfile	56	Generate list file

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Features

The PKZIP 2 software utilities include many features. Here are just a few of them:

Improved Compression. PKZIP 2 implements a new compression algorithm called Deflating. Deflating has varying levels of compression and speed available allowing you to specify the method of compression to be employed. Both the speed and amount of compression are improved over previous versions of PKZIP.

Multi-volume Archive Support. PKZIP 2 can create .ZIP files larger than a single floppy disk, and it can format the floppy disk on the fly!

Subdirectory Storage. PKZIP 2 can find and store subdirectory pathnames within a .ZIP file. Files can then be restored to their original subdirectories, or they may be re-created. Storage includes empty subdirectories.

Automatic Detection and Utilization of 80386 and 80486 CPU's, EMS and XMS Memory, Novell Netware, and 32-bit DPML. PKZIP 2 gets the most out of your machine by using the power of modern CPU's. EMS and XMS support offer the ability to process more files with less available conventional memory. Novell Netware and 32-bit DPML support offer speed improvements.

The above feature also allows you to get the best performance when in a DOS session under OS/2 or Windows. OS/2 and Windows offer EMS, XMS and 32-bit DPML without your using extra drivers.

Memory Requirements. PKZIP requires DOS 2.0 or above and a minimum of 206k of conventional memory. If EMS is available, the conventional memory requirement is 108k. PKUNZIP requires 90k of conventional memory, PKUNZIP JR requires about 30k and PKSFX requires a minimum of 71k.

Easy to use Configuration Program. PKZIP 2 can be configured to best suit your needs through the use of the PKCFG program.

PKSFX Junior and PKUNZIP Junior. These "Junior" programs use very little memory and are extremely small. Although they are reduced in ability, their small size makes them very useful.

Password Encryption. Sensitive data files can be scrambled with password protection. The security of this feature has been improved in PKZIP 2.

Authenticity Verification. Users requiring authentication of archive files may request an Authorization Verification Code. This allows users to create .ZIP files which reveal tampering.

Introduction

Welcome to PKZIP 2.0. This release continues the high performance tradition of previous versions of PKZIP, with a host of new features. In addition to an improved reference section, this manual contains a step-by-step tutorial. By following the *Installation*, *Tutorial* and *Advanced Features* sections in order you will learn how to use PKZIP to your best advantage. Use the *Command Reference* sections in your day-to-day use of PKZIP. Convenient indices to these sections are printed on the inside of the front and back covers.

PKZIP is a powerful program. It has many options, but it is possible to make use of PKZIP by knowing only a few simple options. You can start out with a basic understanding and learn more if and when you need to use additional features.

It is recommended that you first read through this introduction and the installation procedure. Then perform the installation and work through the Tutorial in order.

Follow up with the *Advanced Features* section of the Tutorial. You should not attempt to use the *PKZIP Backup Guide* until you have completed (and understand) the entire tutorial.

Conventions

This manual follows several conventions for clarity and to help you find information quickly. Text that you would see on your computer screen is shown here in a “typewriter” typeface. Text that would be entered by you in these examples is shown in bold:

```
C:\DOS>dir
```

Bold text is also used for emphasis and to point out terms which are then shown in vocabulary boxes.

Vocabulary

Terms that are felt to be technical enough to require explanation are listed in convenient vocabulary blocks. These vocabulary terms are with the first instance of the word, so that you may read them and continue on without needing to search for the glossary. This block is an example.



Anything that we want to be sure you don't miss will be pointed out in the same manner as this paragraph.

Command Syntax

Standard command specification syntax is used throughout the *Command Reference* section. This is used to differentiate between command option parameters which are optional or mandatory, and to indicate when certain options may be combined:

- < > Angle Brackets are used to indicate a command option parameter which must be present. The option parameter will follow immediately after the option with no space.
- [] Square Brackets indicate a command option parameter which is not mandatory.
- , | If there are several possible options which may be used in the position indicated by the angle or square brackets, they will be separated by either a Comma or a Vertical Bar. If options are separated by a comma, then more than one of the options may be used simultaneously. If they are separated by a vertical bar, the options are exclusive of one another and only one may be used at a time.

Following are some abstract examples of how these syntax rules are applied.

<u>Syntax</u>	<u>Valid Entries</u>
-z<a,b,c>	-za, -zab, -zac, -zbc, -zabc
-z<a b c>	-za, -zb, -zc
-z[a,b]	-za, -zb, -zab, -z
-z<a b,c>	-za, -zb, -zac, -zbc -zab or -zabc would be invalid.

Product Support

We hope that your use of PKZIP and PKUNZIP will be enjoyable and trouble-free. It is our intention to provide the best possible product and documentation. We understand that problems may arise, and in most cases you will be able to resolve any problems you may have by referring to the documentation. Answers to questions you might have will usually be found within these pages.

You may reach our Product Support staff in several convenient ways.

Support Conferences

PKWARE maintains support conferences on several BBS Networks. Such as ILink, SmartNet and Rime. You may leave a question in the PKWARE support conference on any of these networks. A Support Technician or fellow user will answer your questions. Many other BBS networks also offer PKWARE conferences, but these are not necessarily monitored by PKWARE.

PKWARE BBS

PKWARE also maintains its own BBS. The PKWARE BBS is normally available 24 hours a day. The BBS supports all baud rates from 1200 up to 33.6K. Set your communications software for 8 bits, 1 stop bit, no parity.

The PKWARE BBS offers the latest in PKWARE shareware as well as many other files for downloading. The PKWARE BBS has multiple phone lines for your convenience.

PKWARE BBS
(414)354-8670

CompuServe

PKWARE Support Technicians maintain a discussion forum on CompuServe. To get to the forum, simply type "GO PKWARE" from any CompuServe prompt. PKWARE is in area 11.

PKWARE CompuServe ID
75300,730

Internet/Usenet

PKWARE products are frequently discussed in the newsgroup "comp.compression". Internet newsgroups are not supported directly by PKWARE.

Through Internet

WWW - <http://www.pkware.com>; Email - support@pkware.com

Voice and Fax

You may reach a PKWARE Support Technician by sending a Fax containing your question.

PKWARE FAX
(414)354-8559

If your question is urgent, or you do not have access to any of the above methods of communication, a Support Technician may be reached by phone. PKWARE Technical support is normally available Monday through Friday, 9 am to 5 pm Central Time (-7 hours GMT).

PKWARE Voice Support
(414)354-8699

Information to Give

When contacting PKWARE Technical Support in any of the above methods, please be prepared to provide as much information about the problem as possible. Include the version numbers for: PKZIP, DOS, and any Memory and Device Drivers you are using. Include the command(s) you have tried to use, and what the results were and note why you feel you have a problem.

When faxing a problem, a print out of any error messages is often helpful in determining a solution.

Please see the *Trouble Shooting* section for additional information.

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Installation

PKZIP works best when it is located in your path. You can put PKZIP in your path by either copying it into an already existing subdirectory that is in your path, or by copying it into its own subdirectory and putting that subdirectory into your path.



Throughout this manual many examples will show the command prompt. This will look like "C:\>" or "C:\PKWARE>" or something similar. This represents the command prompt displayed by your computer and is not to be typed in. If you do not see a prompt that looks like this when you are using DOS, try typing the command "PROMPT \$P\$G". This will set your prompt to resemble those used here. You may also want to add the "PROMPT \$P\$G" command to your AUTOEXEC.BAT file.

Path

There are two meanings for path as used under MS-DOS. It may refer to the **location** of a file, or it may refer to the places DOS will search for a command.

In the first instance, a path would be the directory or subdirectories that a file is located in. If a file called "HOURS.PYS" is in a subdirectory called "JUNE" under a directory called "PAYROLL" on the D: hard disk, its full path would be "D:\PAYROLL\JUNE\HOURS.PYS".

In the second instance, the path is a list of places stored in an **environment variable** called "path". If you type "path" at the DOS prompt and press enter you will see the path that is currently set on your computer. It will usually look something like this:

```
C:\;C:\DOS;C:\UTIL;D:\WINDOWS
```

(It will look different on your machine). The path is set by a statement in the AUTOEXEC.BAT file. For more information, please see your DOS manual.

Installing PKZIP On Your Hard Drive

Your computer should be turned on and at the DOS prompt. If your computer normally runs a program automatically when you turn it on, you will need to exit from this program. You should have a prompt on your screen similar to:

```
C:\>
```



This installation example shows PKZIP being run from the C:\PKWARE directory. PKZIP may be run from any drive in any directory.

Place the diskette containing the original PKZIP distribution in your floppy drive, either A: or B:. Next, create a subdirectory for PKZIP on your hard drive. To do so type:

```
C:\>MD \PKWARE
```

Then change to this subdirectory.

```
C:\>CD \PKWARE
```

```
C:\PKWARE>
```

Now enter the name of the self-extractor on your PKZIP disk, include the drive letter as part of the name.

```
C:\PKWARE>A:PKZ200
```



Note that the name of the self-extractor used throughout this manual (PKZ200.EXE) may vary but will follow the pattern of PKZ###.EXE where ### represents the version number of the software.

The contents of this file will now extract into the PKWARE subdirectory.



The A: and C: drives in the above command are only for example, and may be replaced with other drive letters as appropriate to your particular setup.

You must next modify your AUTOEXEC.BAT file. You can edit this file with any editor that can load and save DOS TEXT or plain ASCII files (For example, “edit” in DOS 5.0, Windows Write, or any word processor with a DOS TEXT editing ability. Consult your word processor’s manual if you are unsure).

If you have DOS 5.0 type:

```
C:\>edit autoexec.bat
```

Look for a line in the file that says something similar to:

```
PATH=C:\;C:\DOS;(etc...)
```

Go to the end of this line and add a semicolon if one is not already there, and the drive and path to PKZIP that you just created. For example “...;C:\PKWARE”

The line will now look similar to this:

```
PATH=C:\;C:\DOS;C:\PKWARE
```

You will most likely have additional items between the C:\DOS; and C:\PKWARE.

Next add a line that looks like this:

```
set pkware.ini=c:\pkware
```

The directory specified must be equal to the directory where PKWARE.INI is located. If you have followed this installation procedure exactly, the above line will work fine. This line should be near the top of the AUTOEXEC.BAT file.

Save the file and exit. You will have to re-boot your computer to make the change take effect.

After you reboot type "path" at the command line and press . Your machine will show the path. Make certain that the subdirectory you placed the PKWARE files in is shown in the path.

You may now verify that PKZIP is installed by typing "PKZIP" at the DOS command prompt. After you press  you should see a banner screen from PKZIP. Try PKUNZIP as well. You should be able to type PKUNZIP or PKZIP in any directory and drive, not just in the PKWARE subdirectory.

If you receive the message "Bad command or file name" or "Unknown command", PKZIP is not properly installed. Check the location of the files you extracted from the distribution file. Also check the lines that were added to the AUTOEXEC.BAT file. Make certain you have rebooted since the changes were made to the AUTOEXEC.BAT file.

The Addendum File

In addition to the files that make up the PKZIP program, you also have a text file containing any last minute changes or updates to the documentation. *You need to print the addendum file in order to be aware of anything that is not mentioned in the manual, or has changed since this manual was printed.*

To print the addendum, load the text file ADDENDUM.TXT into your text editor and print it.

If you have Windows you may load and print the file ADDENDUM.TXT from within your Windows Write program.

Tutorial

This section provides both a step-by-step tutorial in the usage of the PKZIP utilities and in-depth explanations of some of the more complex options. By following through the lessons in order, you can gradually increase your understanding of the PKZIP utilities and your computer system.



It is assumed that you have successfully completed the installation of PKZIP and understand basic DOS commands and usage. You must understand and be able to use the following DOS commands: `cd`, `copy`, `del`, `dir`, `md(mkdir)`, `rd(rmdir)`. You should also be familiar with the concepts of wildcards and directory structures. If you are not sure about usage of any of these please review your DOS manual.

Things that will appear on your computer screen are shown in this manual with a special typeface. Text which you type will be shown in bold.

For example:

`C:\>dir`

Wildcards

DOS makes use of two standard wildcard characters, `*` and `?`. These are the any/all and single character wildcards. An `*` may be used to specify any character, any number of characters. For example a directory of `*.exe` would produce a directory listing of ALL `.exe` files regardless of the length or value of their names. The `?` is used to specify any SINGLE character. A directory of `?.exe` would only show `.exe` files with single character names.

This can be very useful if you have a number of files with similar names to operate upon, such as `biscuit1.txt` `biscuit2.txt` etc. All files matching this pattern could be specified as `"biscuit?.txt"`

For additional information please see your DOS manual.

Getting Started

Creating a Workspace

Before we start learning to use PKZIP, create a directory in which to work. This will give you a controlled environment so that you can more closely follow the tutorial and not have to worry about affecting other files.

We will create a tutorial directory and copy some files into the directory so that we can practice manipulating the files with PKZIP and PKUNZIP. This tutorial assumes that you have created the directory on the C: drive, but it could be created on any drive you have.

Your screen should be at the DOS prompt. You will see something like this:

```
C:\>
```

To make certain that you are at the root of your C: drive type the following:

```
C:\>c:
C:\>cd \
```

Now create the tutorial directory:

```
C:\>md ziptut
C:\>cd ziptut
C:\ziptut>
```

If after changing directory (cd) to the tutorial directory your prompt does not appear as it does above (C:\ziptut) type the following:
C:\>prompt \$p\$g

Now we are going to copy the specimen files into the tutorial directory. These files are completely arbitrary. Their names or contents will not greatly affect your ability to follow this tutorial. If you do not have all of the files specified here, you can substitute other file(s).

The files will come from the DOS and PKWARE directories.

```
C:\ziptut>cd \dos
C:\dos>copy ansi.sys c:\ziptut\ansi.sys
        1 file(s) copied
C:\dos>copy tree.com c:\ziptut\tree.com
        1 file(s) copied
C:\dos>copy more.com c:\ziptut\more.com
        1 file(s) copied
C:\dos>copy attrib.exe c:\ziptut\attrib.exe
        1 file(s) copied
C:\dos>copy find.exe c:\ziptut\find.exe
        1 file(s) copied
C:\dos>cd \pkware
C:\pkware>copy appnote.txt c:\ziptut\appnote.txt
        1 file(s) copied
```

Now you can go to the tutorial directory and see what files are there:

```

C:\pkware>cd \ziptut
C:\ziptut>dir

Volume in drive C is MS-DOS_5
Volume Serial Number is 1831-91EF
Directory of C:\ZIPTUT

.                02-13-92  10:59a
..               02-13-92  10:59a
ANSI             SYS      9029  04-09-91   5:00a
TREE            COM      6901  04-09-91   5:00a
MORE            COM      2618  04-09-91   5:00a
ATTRIB          EXE     15796  04-09-91   5:00a
FIND            EXE      6770  04-09-91   5:00a
APPNOTE         TXT     32208  11-26-91   1:31p
               8 file(s)      73322 bytes
               4173824 bytes free

```



File sizes and dates on the files you are using may vary from those shown in the text of this tutorial.

Simple PKZIP and PKUNZIP usage

Now that we have a controlled place to learn, we can get started. Let's try a couple of simple operations. First, let's create a .ZIP file called "test" which contains a copy of all the files in this directory:

```
C:\ziptut>pkzip test
```

That's all you have to type!

```

PKZIP (R)  FAST!  Create/Update Utility  Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745

```

```

■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ XMS version 2.00 detected.

```

```

Creating ZIP: TEST.ZIP
Adding: ANSI.SYS      Deflating (31%), done.
Adding: TREE.COM      Deflating (33%), done.
Adding: MORE.COM      Deflating (22%), done.
Adding: ATTRIB.EXE    Deflating (35%), done.
Adding: FIND.EXE      Deflating (29%), done.
Adding: APPNOTE.TXT   Deflating (68%), done.

```

Why was that so easy? This is the simplest command that can be given to PKZIP. To understand why that worked you must know two things: The first file on the command line is always a .ZIP file. Secondly, if you don't specify what files to ZIP, it will ZIP all files.

What is a .ZIP file?

It's probably easiest to think of a .ZIP file as a sort of envelope for files. When a file is "zipped", a copy of the file is placed into the .ZIP file.

During this “zipping” process, several things are retained. The most important item retained, of course, is the data that makes up the file itself. Usually this copy of the file is compressed.

Additional information stored about each file includes the date and time that appeared in the directory information about the file, the size before and after compression, and the CRC(see below) value of the file. Optional information such as the file attributes and a comment may also be stored.

CRC

This is an acronym for Cyclic Redundancy Check. When a CRC is performed, the data making up a file is passed through an algorithm. The algorithm computes a value based upon the contents. The result is an eight digit hexadecimal number representing the value of the file.

If even a single bit of a file is altered, and the CRC is performed again, the resulting CRC value will be different. By using a CRC value, it can be determined that there is an exact match for a particular file.

PKZIP calculates a CRC value for the original file before it is compressed. This value is then stored with a file in the .ZIP file. When a file is extracted by PKUNZIP or PKSFX, it calculates a CRC value for the extracted data and compares it against the original CRC value. If the data has been damaged or altered, PKUNZIP/PKSFX can recognize and report this.

Viewing a .ZIP file

Let's look at the contents of the .ZIP file we have just made. To do this we will use the view option of PKZIP. PKUNZIP also supports this option. Using this command will give us a sort of “directory” of the .ZIP file.

```
C:\ZIPTUT>pkzip test -v
```

or (both command line styles work the same)

```
C:\ZIPTUT>pkzip -v test
```

```
PKZIP (R) FAST! Create/Update Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

```
■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ XMS version 2.00 detected.
```

(continued next page)

Searching ZIP: TEST.ZIP

Length	Method	Size	Ratio	Date	Time	CRC-32	Attr	Name
9029	DeflatN	6035	34%	04-09-91	05:00	bc7cf24c	--w-	ANSI.SYS
6901	DeflatN	4572	34%	04-09-91	05:00	3c8d7b27	--w-	TREE.COM
2618	DeflatN	1978	25%	04-09-91	05:00	cd42e73d	--w-	MORE.COM
15796	DeflatN	10056	37%	04-09-91	05:00	35522d06	--w-	ATTRIB.EXE
6770	DeflatN	4709	31%	04-09-91	05:00	70cb0ef5	--w-	FIND.EXE
32208	DeflatN	9988	69%	11-26-91	13:31	9933dcd8	--w-	APPNOTE.TXT
73322		37338	50%					6

C:\ZIPTUT>

The information listed includes the file *Name*, *Date* and *Time* as it appears in a normal directory listing. Additional information includes the *Length* (normal file size), the *Method* of compression used, and its current *Size* (compressed).

The *Ratio* is a measure of how much space was saved. It is the percent of size removed from the original file length. A 100k file that is reduced to 25k would have a Ratio of 75%.

The *CRC-32* is the CRC value computed when the file was compressed. Note that the CRC value is independent of the type of compression used. The *Attr* (Attribute) column contains file attribute information.

Zippping a Single File

Now let's create a .ZIP file which contains only a single file. Issue the following command:

C:\ZIPTUT>**pkzip test2.zip appnote.txt**

Since the first filename on the PKZIP or PKUNZIP command line is always a .ZIP file, the ".ZIP" at the end of the filename is optional. You could simply type:

C:\ZIPTUT>**pkzip test2 appnote.txt**

You have now created a .ZIP file containing only a single file. Try viewing it just as you viewed the first .ZIP file you created. Type **dir** to get a directory of the ZIPTUT directory. Notice that you still have the original six files plus two .ZIP files. The files that you Zipped are still there.

Moving Files

Good computing practice dictates that you should make a backup copy of files before you delete the original copy.

What if you want to not just copy the files, but want to actually move them into a .ZIP file? By adding an option to the command line when you ZIP the file, you can have the file moved from the original location into the .ZIP file. This saves the space by which the file is reduced. For example, a 100k file reduced to a 30k .ZIP file saves 70k.

Let's try it with the file appnote.txt. Type this command:

```
C:\ZIPTUT>pkzip test3 appnote.txt -m
```

The new option used here is -m, for move. Now perform a directory listing (**dir** ). Notice that you have created a third .ZIP file, and that the file appnote.txt is gone. It now exists only in the .ZIP file.

Let's take the file back out (extract):

```
C:\ZIPTUT>pkunzip test3
```

Like PKZIP, the default, if no files are specified, is to extract all of them. Since the .ZIP file "test3" contains only appnote.txt, this is the only file that will be extracted.

Now extract the first .ZIP file we created:

```
C:\ZIPTUT>pkunzip test
```

```
PKUNZIP (R)  FAST!  Extract Utility  Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

```
Searching ZIP: TEST.ZIP
```

```
PKUNZIP: (W18) Warning! ANSI.SYS already exists. Overwrite (y/n/a/r)?a 
Inflating: ANSI.SYS
Inflating: TREE.COM
Inflating: MORE.COM
Inflating: ATTRIB.EXE
Inflating: FIND.EXE
Inflating: APPNOTE.TXT
```

```
C:\ZIPTUT>
```

What has happened here? When the file was extracted this time, there were already files in the directory with similar names. Before PKUNZIP extracts a file, it checks to see if a file with the same name already exists. If the file already exists, PKUNZIP will prompt you. This is to help you avoid unintentionally overwriting a file.

After warning you and listing the file name, PKUNZIP will ask:
Overwrite (y/n/a/r)? This means, should the original file be overwritten with the one in the .ZIP file? You can answer: Yes, No, All or Rename.

If you answer Yes to this question, PKUNZIP will overwrite the file in the directory with the one contained in the .ZIP file. If you answer No, it will skip this file and go on to the next file(if any).

Answering "All" means Yes, overwrite this file, and overwrite any others as well. This would be the same as answering Yes to each individual file that is found to already exist. The command line option, -o, will do the same thing without any prompting.

Rename will allow you to extract this file and not overwrite the original one. You will be prompted by PKUNZIP for a new filename to use. Feel free to experiment with this.

Zippping Multiple Files

Now we'll ZIP several files at once, but not all files, as was done in the previous examples.

Suppose we want to create a .ZIP file which contains all the files in this directory that have an extension of .COM. To do so, we use standard DOS wildcards when specifying the file name:

```
C:\ZIPTUT>pkzip test4 *.com

Creating ZIP: TEST4.ZIP
Adding: TREE.COM Deflating (34%), done.
Adding: MORE.COM Deflating (25%), done.
```

Note that you can combine options. If you wanted to move these files while creating the .ZIP file, you would issue the command as:

```
C:\ZIPTUT>pkzip test4 *.com -m
```

Extract these files before continuing:

```
C:\ZIPTUT>pkunzip test4
```

Now create another .ZIP file. This one will have all files except the files ending in .COM.

```
C:\ZIPTUT>pkzip test5 -x*.com
```

Here, you aren't telling PKZIP what files to take, so it assumes that it should take everything. You are also telling PKZIP which file(s) not to take. The -x option means to exclude. When you try this you will see that PKZIP is taking copies of the .ZIP files too. You can exclude .ZIP files at the same time:

```
C:\ZIPTUT>pkzip test5 -x*.com -x*.zip
```

It is acceptable to exclude two or more types of files.

You may have as many different file patterns on the command line as will fit. This includes filenames, file patterns, and exclusions.

We seem to have quite a few .ZIP files now. Let's delete them before continuing:

```
C:\ZIPTUT>del *.zip
```

And create a single .ZIP file to work with again:

```
C:\ZIPTUT>pkzip fubar.zip
```

Testing .ZIP files

Before putting an important .ZIP file away for storage, or sending it to another person, it is a good idea to test it. This will verify that the .ZIP file created is not damaged, and that the area of the disk it is located on has no errors. For this purpose there is a PKUNZIP option called test.

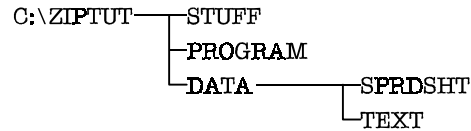
```
C:\ZIPTUT>pkunzip fubar -t
```

```
PKUNZIP (R)  FAST!  Extract Utility  Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

```
Searching ZIP: FOOBAR.ZIP
Testing: ANSI.SYS      OK
Testing: TREE.COM      OK
Testing: MORE.COM      OK
Testing: ATTRIB.EXE    OK
Testing: FIND.EXE      OK
Testing: APPNOTE.TXT   OK
```

This is what a successful test of a .ZIP file looks like. As each file is tested, "OK" is displayed next to the filename.

Storing and Rebuilding Directory Structures



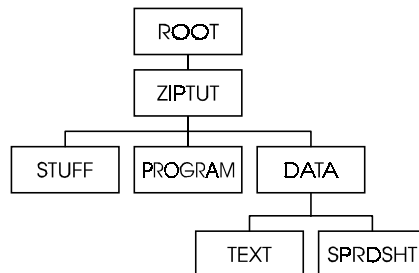
Above is a directory structure which you are going to create in order to learn how to store directories. Issue the following series of commands:

```
C:\ZIPTUT>md stuff
C:\ZIPTUT>md program
C:\ZIPTUT>md data
C:\ZIPTUT>cd data
C:\ZIPTUT\DATA>md sprdsht
C:\ZIPTUT\DATA>md text
C:\ZIPTUT\DATA>cd ..
```

Next you need to put at least one file in each of these subdirectories.

The following copy commands would accomplish this:

```
C:\ZIPTUT>copy ansi.sys data\text
1 file(s) copied
C:\ZIPTUT>copy tree.com data\sprdsht
1 file(s) copied
C:\ZIPTUT>copy more.com stuff
1 file(s) copied
C:\ZIPTUT>copy attrib.exe program
1 file(s) copied
C:\ZIPTUT>copy appnote.txt data
1 file(s) copied
```



The Original Tree Structure

Now try this command. Make certain you are in the ZIPTUT directory!

```
C:\ZIPTUT>pkzip pathtest -rp
```

PKZIP (R) FAST! Create/Update Utility Version 2
 Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
 PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745

- 80486 CPU detected.
- EMS version 4.00 detected.
- XMS version 2.00 detected.
- Using Normal Compression.

```

Creating ZIP: PATHTEST.ZIP
Adding: ANSI.SYS           Deflating (34%), done.
Adding: TREE.COM           Deflating (34%), done.
Adding: MORE.COM           Deflating (25%), done.
Adding: ATTRIB.EXE         Deflating (37%), done.
Adding: FIND.EXE           Deflating (31%), done.
Adding: APPNOTE.TXT         Deflating (69%), done.
Adding: STUFF/              storing ( 0%), done.
Adding: PROGRAM/            storing ( 0%), done.
Adding: DATA/              storing ( 0%), done.
Adding: STUFF/MORE.COM      Deflating (25%), done.
Adding: PROGRAM/ATTRIB.EXE Deflating (37%), done.
Adding: DATA/SPRDSHT/      storing ( 0%), done.
Adding: DATA/TEXT/         storing ( 0%), done.
Adding: DATA/APPNOTE.TXT   Deflating (69%), done.
Adding: DATA/SPRDSHT/TREE.COM Deflating (34%), done.
Adding: DATA/TEXT/ANSI.SYS Deflating (34%), done.
  
```

As the example shows, PKZIP not only took all the files as it did before, but now it also Zipped up all of the subdirectories. Notice that the path is stored (the subdirectory name) with each file. View the file with the -v option. Notice how the file names appear.

C:\ZIPTUT>pkunzip pathtest -v

PKUNZIP (R) FAST! Extract Utility Version 2
 Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
 PKUNZIP Reg. U.S. Pat. and Tm. Off.

Searching ZIP: PATHTEST.ZIP

Length	Method	Size	Ratio	Date	Time	CRC-32	Attr	Name
9029	DeflatN	6035	34%	04-09-91	05:00	bc7cf24c	--w-	ANSI.SYS
6901	DeflatN	4572	34%	04-09-91	05:00	3c8d7b27	--w-	TREE.COM
2618	DeflatN	1978	25%	04-09-91	05:00	cd42e73d	--w-	MORE.COM
15796	DeflatN	10056	37%	04-09-91	05:00	35522d06	--w-	ATTRIB.EXE
6770	DeflatN	4709	31%	04-09-91	05:00	70cb0ef5	--w-	FIND.EXE
32208	DeflatN	9988	69%	11-26-91	13:31	9933dcd8	--w-	APPNOTE.TXT
0	Stored	0	0%	03-23-92	11:39	00000000	--wD	STUFF/
0	Stored	0	0%	03-23-92	11:39	00000000	--wD	PROGRAM/
0	Stored	0	0%	03-23-92	11:39	00000000	--wD	DATA/
2618	DeflatN	1978	25%	04-09-91	05:00	cd42e73d	--w-	STUFF/MORE.COM
15796	DeflatN	10056	37%	04-09-91	05:00	35522d06	--w-	PROGRAM/ATTRIB.EXE
0	Stored	0	0%	03-23-92	11:39	00000000	--wD	DATA/SPRDSHT/
0	Stored	0	0%	03-23-92	11:39	00000000	--wD	DATA/TEXT/
32208	DeflatN	9988	69%	11-26-91	13:31	9933dcd8	--w-	DATA/APPNOTE.TXT
6901	DeflatN	4572	34%	04-09-91	05:00	3c8d7b27	--w-	DATA/SPRDSHT/TREE.COM
9029	DeflatN	6035	34%	04-09-91	05:00	bc7cf24c	--w-	DATA/TEXT/ANSI.SYS
139874		69967	50%					16

Now you can try this. Go to the root of the drive.

C:\ZIPTUT>cd\

Then create another tutorial directory.

```
C:\>md ziptut2
C:\>cd ziptut2
```

Now extract the file into this new directory.

```
C:\ziptut2>pkunzip -d c:\ziptut\pathtest
PKUNZIP (R) FAST! Extract Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

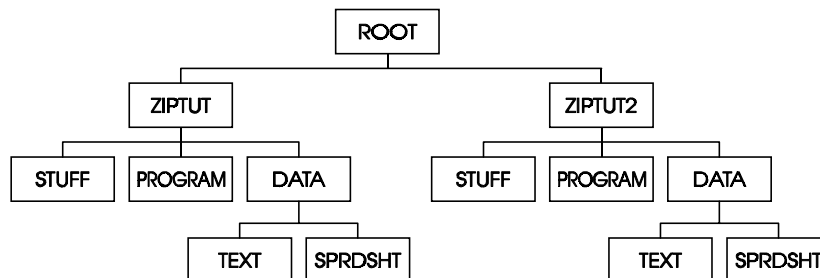
```
Searching ZIP: c:/ziptut/PATHTEST.ZIP
Inflating: ANSI.SYS
Inflating: TREE.COM
Inflating: MORE.COM
Inflating: ATTRIB.EXE
Inflating: FIND.EXE
Inflating: APPNOTE.TXT
Extracting: STUFF/
Extracting: PROGRAM/
Extracting: DATA/
Inflating: STUFF/MORE.COM
Inflating: PROGRAM/ATTRIB.EXE
Extracting: DATA/SPRDSHT/
Extracting: DATA/TEXT/
Inflating: DATA/APPNOTE.TXT
Inflating: DATA/SPRDSHT/TREE.COM
Inflating: DATA/TEXT/ANSI.SYS
```

All of the files and paths are re-created.

If you have the DOS tree command available, you can try this:

```
C:\ZIPTUT2>tree
Directory PATH listing for Volume MS-DOS_5
Volume Serial Number is 1831-91EF
C:.\
+----STUFF
+----PROGRAM
+----DATA
+-----SPRDSHT
+-----TEXT
```

The tree structure now looks like this:



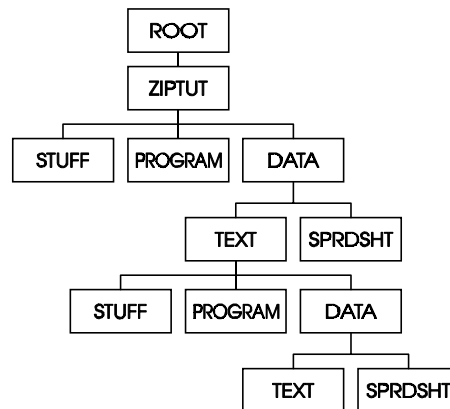
Notice that PKUNZIP will reconstruct the file structure in any location. This makes it easier to move directory structures and files around, from drive to drive or machine to machine.

For example, we could have extracted the above .ZIP file on a different drive. Suppose you decided to create a directory called TUTOR to hold a number of tutorials you were working on, and you wanted to put the PKZIP tutorial in a subdirectory. By creating a .ZIP file and extracting it to the new directory you could create a new copy in the desired directory.

Before we continue, let's clean up. You can easily delete the second copy of the ZIPTUT directory with PKZIP. Create a .ZIP file of the directory structure, and tell PKZIP to move the files with the -m option. Then delete the JUNK.ZIP file.

```
C:\ZIPTUT2>cd\  
C:\>pkzip -rpm junk.zip ziptut2\*.*  
C:\>del junk.zip
```

If you were to specify the target directory as one of the original subdirectories you could make the directory structure look like this:



The above structure was created by changing to the directory

C:\ziptut\data\text and issuing the command:

```
C:\ZIPTUT\DATA\TEXT>pkunzip -d c:\ziptut\pathtest.zip
```

Deleting Files from a .ZIP file

Now that you have all these files in a .ZIP file, what if you want to delete something from the .ZIP file? You use PKZIP to delete files from a .ZIP file. The option used is -d which means delete. The command structure is simple and very similar to that used when adding a file. To delete the file "more.com" from the pathtest.zip file:

```
C:\ZIPTUT>pkzip pathtest -d more.com
```

Go ahead and try it now. You can not only delete specific files, but you can also delete files matching a pattern:

```
C:\ZIPTUT>pkzip pathtest -d *.exe
```

```
PKZIP (R)  FAST!  Create/Update Utility  Version 2
Copr. 1989-1992 PKWARE Inc.  All Rights Reserved.  PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off.  Patent No. 5,051,745
```

```
Updating ZIP: PATHTEST.ZIP
Deleting: ATTRIB.EXE
Deleting: FIND.EXE
Deleting: PROGRAM/ATTRIB.EXE
```

Notice that when you deleted “more.com” previously, only the copy in the “root” of the .ZIP file was deleted, not the copy in the “stuff” directory. PKZIP/PKUNZIP see these files as being different, this will prevent making mistakes by deleting the wrong file. You must specify the exact file you want to delete, including the stored path name.

You may delete files from a directory by pattern/filespec as well:

```
C:\ZIPTUT>pkzip pathtest -d data\*.*
```

```
PKZIP (R)  FAST!  Create/Update Utility  Version 2
Copr. 1989-1992 PKWARE Inc.  All Rights Reserved.  PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off.  Patent No. 5,051,745
```

```
Updating ZIP: PATHTEST.ZIP
Deleting: DATA/
Deleting: DATA/SPRDSHT/
Deleting: DATA/TEXT/
Deleting: DATA/APPNOTE.TXT
Deleting: DATA/SPRDSHT/TREE.COM
Deleting: DATA/TEXT/ANSI.SYS
```

Speed vs. Size

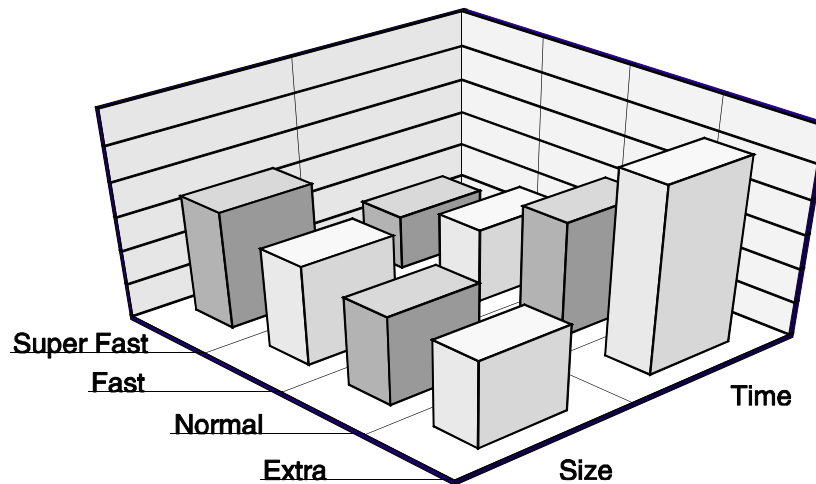
PKZIP offers a number of choices as to how fast it will compress and how small the final .ZIP file will be. For some users of PKZIP compression is most important. Other people are not concerned as much with the size of the .ZIP file, but rather with how long it will take to process the files.

The default compression method of PKZIP is the best balance between compression speed and size.

The four levels of compression and speed offered by PKZIP are referred to as “Maximum”, “Normal”, “Fast” and “Super Fast”.

In the graph below, the bars at the back right represent the relative time it takes to compress the files. The bars on the front left represent the relative size of the files after compression. The bars are in pairs of the same color, with the time being at the left front and the file size to the right and back. The values of the graph are based on actual data but are represented here in an arbitrary fashion to demonstrate the concept of *Speed vs. Size*.

As you can see, the time varies in a regular manner, while the file size remains fairly similar until the fastest method is used. Use this graph and your own experience with PKZIP 2 to determine what level fits your everyday needs.

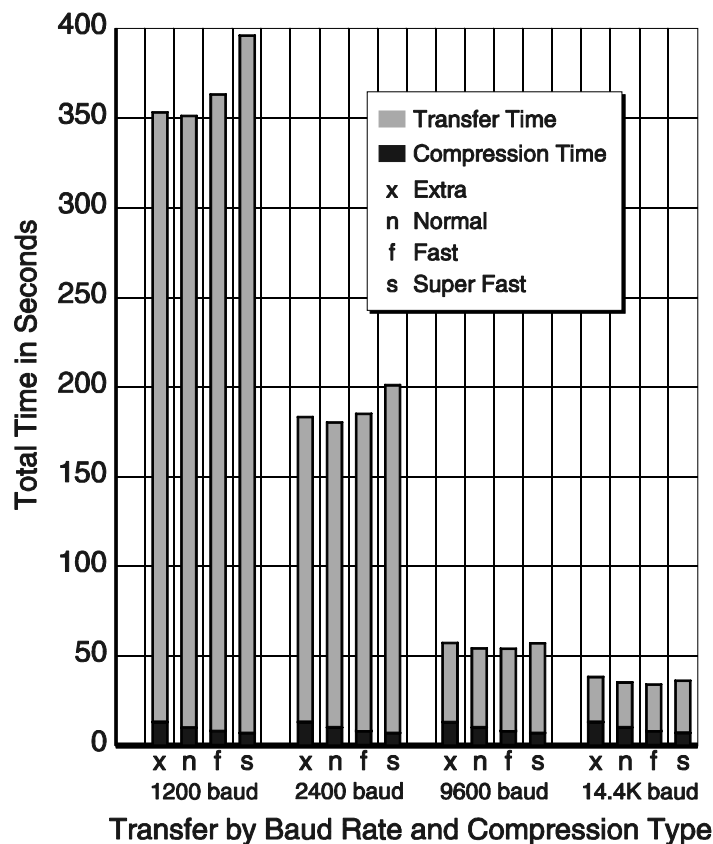


This graph shows the relative file size and compression time resulting from the four available compression methods.

Other Factors in Speed vs. Size

This is a practical example of why the compression method you choose can be important. Many BBS's use files called ".QWK" (quick) packets to transfer messages back and forth. While a user is logged into the BBS, a program takes all of the messages that are new and places them into a .ZIP file. The .ZIP file is then downloaded to be read on the user's machine.

This is an instance where both the compression time and the compression size are important. The user is making a phone call during this entire time, and must wait for both the compression and for the file to be transferred.



The graph shows the file transfer time stacked on top of the compression time. The resulting column height represents the total time (in seconds) involved when receiving a .QWK packet. The .QWK packet is approximately 110K and was compressed on a 486/33. Note that larger packets and slower CPU's will result in greater time differences between compression methods.

In this case, the usual assumption of wanting best speed or best compression is shown to be false. Choosing the fastest compression will yield a larger file, which will take more time to transfer. Choosing the maximum amount of compression will take more time than is warranted by the reduced filesize.

From the chart it is evident that when transmitting a QWK packet at high baud rates the “fast” method is best, while at low baud rates the default speed is best.

Selecting the Compression Type

The compression method is selected with the -e option. Follow the -e with the letter corresponding to the compression method you wish to use.

The letters corresponding to the methods are:

- X Extra or maximum compression
- N Normal compression (default)
- F Fast compression
- S Super Fast compression

If you do not specify what type of compression to use, PKZIP will default to “normal” compression.

```
C:\>pkzip -ex jones.zip c:\katie\*.*
```

The -e option is used in conjunction with any other options on a normal command line:

```
D:\>pkzip -rp -es z:\backups\jjp_d.zip *.*
```

PKZIP will report the type of compression it is using in the banner. For example:

- Using Maximum Compression.
- Using Normal Compression.
- Using Fast Compression.
- Using Super Fast Compression.

You may set up your copy of PKZIP so that it defaults to a compression method other than “normal”. This is done with an option in the PKZIP.CFG file. For more information see the PKZIP Configuration section.

Using Advanced Features

This section covers features which are considered to be “advanced”. Please make certain that you are familiar with the features covered in the first section before starting this section.

Encryption

For an explanation of the limitations, implications and use of PKZIP data encryption, please read the separate section *Using Data Encryption*.

Extracting Files to a Printer

It is possible to send a file directly from the .ZIP file to the printer. This is useful for printing documentation or graphic dump files.

There are two important parts to this command. The first specifies whether the file being extracted (and printed) is an ASCII or Binary file. You should consider plain text files such as “.DOC” and “.TXT” files to be ASCII. Files which are re-directed printer dump files should be considered Binary.

Printer Dump

Under MS Windows and some other environments, you have the ability to re-direct printer output to a file. This means that the instructions which would normally be sent to the printer, describing the page to be printed, will be collected into a file.

This file can then be sent to a printer at a later time, producing the document as though it had been printed from the application.

These printer dump files are often large, as they may contain not only text information but also fonts, graphics and bitmaps.

The second part of the command specifies where the file should go. The file may be directed to any parallel or serial port.

This is a practical example. This command will extract the application notes from the PKZIP distribution file and send them to the default printer (PRN:).

```
C:\PKWARE>pkunzip pkz2??.exe appnote.txt -p
```

As you can see, all that was needed was the -p (extract to printer). This is because the default is extraction to PRN:.

Here is a fictional example. This would be the extraction of a printer dump file, to a printer attached to the second serial port (COM2).

```
C:\ARCHIVE>pkunzip parts5.zip clutch.prn -pbc2
```

Extracting Files to the Console

Extracting files to the console (screen) is similar to extracting files to the printer. This is only useful for files that contain plain readable ASCII text. Files which have binary content or formatting codes (such as output from a wordprocessor), cannot be displayed properly.

Extracted text is sent directly to the screen, with no intermediate file being created. This can be useful for checking the contents of a file before deciding if you want to extract or print it.

To view the application notes from PKZIP you would use the command:

```
C:\PKWARE>pkunzip pkz2??.exe appnote.txt -cm
```

By using the “m”, you are instructing PKZIP to pause after each screen of text. If the “m” is left off, the text will flow by very quickly.

Storing and Restoring the Volume Label

In some situations the volume label of a diskette may be important. PKZIP offers the ability to store this label as part of a .ZIP file, and then restore it when the file is extracted.

Volume Label

Every drive that can be accessed on your computer is referred to as a VOLUME. This includes not only your hard disk but also individual floppy disks. Each of these drives may have a VOLUME LABEL. This is a text string which identifies the volume. This is particularly useful on removable media, such as floppy disks. Each floppy may have a label which is meaningful to you, or to a piece of software. For further information on VOLUME LABELS please refer to your DOS manual.

Let's assume we have a disk in drive A: with a volume label. We want to ZIP the entire contents of the drive, including subdirectories and any volume label. This is the command that would be used:

```
C:\>pkzip -rp -$a floppy.zip a:*.*
```

The -\$ option tells PKZIP to store the volume label. The letter corresponding to the drive with the volume label should follow. The letter identifying the drive is used to indicate from what drive the volume label will be coming. This is required because it is possible to .ZIP files from multiple drives with one command.

To restore this volume label when the file is extracted, you would simply include the -\$. The volume label will be restored to the target drive.

```
C:\>pkunzip -$ floppy.zip a:
```

Compressing and Extracting Hidden and System Files

PKZIP can optionally store files with the hidden and system file attributes. During normal operation PKZIP will skip any hidden or system files.

File Attributes

Every file on an MS-DOS partition has several properties associated with it. These attributes are: archive(**a**), hidden(**h**), read-only(**r**), and system(**s**). The Archive attribute is explained under the **-i** option. The Hidden attribute prevents MS-DOS from displaying the file in a directory listing. Files that have this attribute are normally ignored by PKZIP. The Read-Only attribute prevents a file from being changed or deleted. When a file has this attribute you cannot delete the file or change its contents. The System attribute signifies a file that is a system file. System files are used by MS-DOS and are treated as Hidden files. Please see your DOS manual for additional information.

To instruct PKZIP to take files which have either the hidden or system attributes set, the **-w** option is used. A letter corresponding to hidden and/or system files then follows.

To compress all files on a disk with this option, the command would be similar to:

```
C:\>pkzip -whs hiddtest.zip a:*.*
```

When PKZIP archives these files, it keeps track of the attributes that identified them as hidden and system. If you do not wish to retain these attributes you must also use the **-j** (lower case) option.



The **-jr** option is used to instruct PKZIP to not retain the read-only attribute of files.

The **-j** option is followed by a "h" or "s" in a manner identical to the **-w** option. This command example would retain the hidden and system file attributes on any files it compressed which had those attributes.

```
C:\>pkzip -whs -jhs hiddtest.zip a:*.*
```

When you extract files, PKUNZIP will not normally retain these attributes. This is for your protection. If PKUNZIP did not strip off these attributes, it would be possible for you to end up with hidden files on your drive when you extracted files provided to you by someone else.

To instruct PKUNZIP to extract these files with the attributes intact, use the **-J** (upper case) option in a manner identical to that of PKZIP:

```
C:\>pkunzip a:testfile.zip -Jhrs
```

Note that you cannot extract a file over an existing read only file. The existing file must first have the read-only attribute removed. This is done with the following command:

```
C:\>attrib -r filename.ext
```

Zipfile and File Comments

.ZIP Files may have two types of comments attached to them. The first type of comment is attached to the .ZIP file itself. The second type can be attached to each individual file within the .ZIP file.

File Comments may be added when files are added to an archive, or they may be added to existing files that are within a .ZIP file. To add comments to files in an existing .ZIP file you would use a command as follows:

```
C:\ZIPTUT>pkzip pathtest.zip -c
```

```
PKZIP (R) FAST! Create/Update Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

```
■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ Novell Netware version 3.11 detected.
```

```
Updating ZIP: PATHTEST.ZIP
ANSI.SYS
Old Comment:
New Comment? ANSI Driver, loaded in config.sys
TREE.COM
Old Comment:
New Comment? Program to display directory structure
```

As each file comment is updated, any existing file comment will be displayed. To leave a file comment unchanged, simply press . To add or change a comment, type your new comment at the prompt and press . Each individual comment is limited to 59 characters in length.

If you are adding files to an existing .ZIP file, you have two options. The first is to add comments for files that are being added to the .ZIP file, as well as being prompted to add/edit comments for files already within the .ZIP file. This is done with the -ac option.

The alternative method is to be prompted for comments only for those files which are being added. By using the -C (upper case) option you will not be prompted to add or edit comments for existing files.

File comments are shown by PKZIP only when a view with comments is done (-vc or -vbc).

```
C:\ZIPTUT>pkzip -vc pathtest
```

Searching ZIP: PATHTEST.ZIP

Length	Method	Size	Ratio	Date	Time	CRC-32	Attr	Name
6901	DeflatN	4572	34%	04-09-91	05:00	3c8d7b27	--w-	TREE.COM
Comment: Program to display file directory structure								
9029	DeflatN	6035	34%	04-09-91	05:00	bc7cf24c	--w-	ANSI.SYS
Comment: ANSI Driver, loaded in config.sys								

PKUNZIP is not capable of displaying individual file comments.

.ZIP file comments, unlike file comments, are displayed whenever a file is extracted or tested. Notice the line "Searching (filename).ZIP -" below. The text following the filename and hyphen is a .ZIP file comment.

+

Q:\GAMES>pkunzip -t #1wolf3d.zip

Searching ZIP: #1WOLF3D.ZIP - PKWARE Support BBS (414)354-8670
Testing: CATALOG.EXE OK
Testing: ORDER.FRM OK
Testing: W3D1-ASP.EXE OK

A .ZIP file comment is added in a manner very similar to that of a file comment. This is an example showing the .ZIP file comment of the above file being changed. Note that the .ZIP file "#1wolf3d.zip" is assumed to already exist in this example:

F:\>pkzip -z #1wolf3d.zip

PKZIP (R) FAST! Create/Update Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745

■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ XMS version 2.00 detected.

Updating ZIP: #1WOLF3D.ZIP - PKWARE Support BBS (414)354-8670
Zip comment ? **This is a new .ZIP file comment!**

.ZIP file comments entered in this manner are limited to a maximum length of 127 characters. The *PKZIP Command Reference* section details a method that allows much larger .ZIP file comments.

File comments and .ZIP file comments can be very useful in several ways. When you are archiving files or sending files to another person, these comments will aid in identifying the .ZIP file contents. It may seem obvious to you what a file contains now, but what will the file name "JANX4T-5.WKS" mean to you in six months?

Congratulations

Now that you have completed this simple tutorial you are ready to try PKZIP on your own. It is suggested that you continue to learn

about PKZIP by reading through the remainder of this manual, and the *PKZIP Backup Guide*.

Suggested Reading

We recommend the following books as being helpful in furthering your understanding of your computer and data compression.

Data Compression:

Nelson, Mark. The Data Compression Book. Redwood City: M&T Books, 1991.

Held, Gilbert. Data Compression: techniques and applications. Great Britain: John Wiley & Sons Ltd., 1983.

DOS/computer use:

Gookin, Dan. DOS for Dummies. San Mateo: IDG Books, 1991.

The Waite Group. Understanding MS-DOS. Carmel: Howard W. Sams & Company, 1989.

QUE Corporation. MS-DOS User's Guide. Carmel: Que Corporation, 1989.

Schidt, Herbert. DOS Made Easy. Berkeley: Osborne McGraw-Hill, 1988.

Many community colleges and universities offer short courses to the community on a variety of computer related subjects. Contact your local educational institution for information.

PKZIP

Command Reference

Items contained in square brackets([]) are optional parameters to an option and need not be included. Where there are two (or more) items separated by a comma, you may use any ONE of those options following the main option character. If there are parameters in angle brackets(<>) you must include that option, or choose one of those in the series. When choosing options from a bracketed series, you may use more than one of these options if they are separated by a comma. If the options are separated by a vertical bar (|) you may only choose one option. See page 2 for examples.

Configurable Items

Options which can be set in the configuration file are marked with this symbol . See the *PKZIP Configuration* chapter for more details.

Command Option Guide

-a+ Turn off archive attribute of added files.

By using this option you can create a complete backup of your disk, while clearing the archive attributes to make the way for incremental archiving.

Incremental archiving makes use of the archive attribute to take only the files which have been modified since the last backup. In order for this process to work smoothly you must first have a complete backup and a clearing of the archive attribute for all files.

```
C:>pkzip f:backup.zip -a+ -rp
```

-b<drive:path> Create temporary .ZIP file on alternate drive

Every time a .ZIP file is updated, PKZIP creates a temporary work file. When the modifications to the .ZIP file are successfully completed, the original .ZIP file is replaced with the temporary work file. This means you must have at least as much additional disk space available as was used by the original .ZIP file.

For Example: If you have an existing .ZIP file of 500K, and you are adding another file to it that takes 10K compressed, you need additional workspace of at least 510K during the update process.

This command option will allow you to create the temporary .ZIP file on a drive different from the one on which the original .ZIP file resides. This allows you to update large .ZIP files when space is limited, such as a large .ZIP file on a floppy disk. Also, by setting this temporary drive to point to a RAM drive, you can speed up the operation of PKZIP.

Immediately following the -b option place the drive and/or path you wish to use for the temporary work file.

```
A:>pkzip test.zip -bc: order.doc
```

```
C:>pkzip wp.zip -bz:\public wp51\*.*
```



It is necessary to specify a path in addition to the drive letter only if you are in a situation where disk space or access is being limited by subdirectory, such as on a local area network.

-
- c Create/Edit file comments for all files
 - ac Create/Edit file comments for all files during update
 - C Add comments for new files only

Each individual file contained within a .ZIP file may have a comment attached to it. Comments are useful for identifying files, especially if you need to go back months later to look at a file, or if you are sending the file to someone else. Each comment may be up to 59 characters long, including spaces and punctuation.

The -c (lower case) option is used when you want to add or edit file comments for all files within an existing .ZIP file.

The -ac option may be used during the .ZIP file creation process, or during an update/freshen process. This will add comments for new files as well as allowing you to edit/add comments for files that were already in the .ZIP file.

```
pkzip test.zip -ac license.doc
```

When each file is added to the .ZIP file you will be prompted for a comment. Type a comment of up to 59 characters, and press . If you are updating an existing .ZIP file and there are files which already have comments, the comment will be displayed on the prompt line for you to edit. You may make changes or simply press to leave it unchanged.

The -C (upper case) option is used when you want to add a file comment to newfiles that are being added to an existing .ZIP file. Files which are not being added, or files which already existed in the .ZIP file will not be prompted for comments.

```
pkzip test.zip -C *.doc
```

In this situation, you would be adding additional files to the file called test.zip. Since the file license.doc was added to it in the previous example, you would not be prompted for a comment for this file.

-d Delete files

This command will allow you to remove individual files from within a .ZIP file. This command is only applicable to already existing .ZIP files.

```
pkzip test.zip -d license.doc
```

In the above example the file license.doc would be removed from the .ZIP file test.zip (assuming that it was in the .ZIP file in the first place).



The deletion of files from within a .ZIP file causes a situation where a temporary .ZIP file will exist during the deletion process. In order to delete a file from a .ZIP file there must be as much disk space free as will be occupied by the modified .ZIP file.

Multiple files may be deleted at one time. You may do this either by placing multiple file names on the command line, or by specifying a list file containing the names of files you want deleted. (See *Using List Files*)

```
pkzip july.zip -d file1.wk file2.wk1
```

```
pkzip tue_back.zip -d @old.lst
```



-e[x|n|f|s] Specify compression method

- ex** Maximum compression (most compression, most time)
- en** Normal compression (default)
- ef** Fast compression
- es** Super Fast compression (least compression, least time)
- e0** Zero compression (storing)

PKZIP 2 provides four different methods of compression, allowing you to choose the compression that best suits your need. You can choose speed, high compression or a balance. The default method will provide good compression at a reasonable speed.

The switches for this option are intended to be mnemonic in nature:

-ex	eXtra or maXimum
-en	Normal
-ef	Fast
-es	Speed/Super Fast
-e0	Zero compression

The only two that may be difficult to remember are the order of Fast and Super Fast, you might want to think “Extra Speed” to keep it clear in your mind.

If you need to complete the compression in the minimum amount of time possible, then choose the fast compression method. To force PKZIP to use its fastest compression method, use the option -es (Super Fast).

To achieve the smallest files possible, specify the maximum compression method using the option -ex (Maximum).

```
pkzip -ex test.zip d:\wp51\*.wp
```

Using no compression with the zero option is useful when archiving files which will either not compress or compress very little. This would generally be true when archiving .ZIP files. By specifying no compression PKZIP will store the files directly rather than first attempting to compress them.

The compression method to use by default can be specified in the configuration file. If you have specified the super fast, fast or maximum compression method in your configuration you may override this and use the normal method by using the option -en.

-f Freshen files in a .ZIP file

When this command is specified, PKZIP will compare the contents of the .ZIP file specified on the command line against the files specified for compression. If a file ALREADY EXISTS in the .ZIP file, PKZIP will check to see if the file exists in the source area (i.e. the files specified on the command line). If the file does exist, it then checks to see if the file in the source area is more recent than the copy contained in the .ZIP file. If it is more recent, it will replace the compressed file with the one found in the source area.

Using this parameter will normally speed up the ZIP process considerably, as PKZIP will be able to skip over files that it has already compressed.

```
pkzip -f baka.zip c:\ranma\*.*
```

This example checks the contents of the .ZIP file “baka.zip” against the files specified by “c:\ranma*.*”. Any file that matches will be

updated if the file in the source area is newer than the copy within the .ZIP file.

See Also: -u, -i

-h Display command summary (Help)

This option will display a brief summary of the command switches available.

-i Add files that have the archive attribute - then turn attribute off

-i- Add files that have the archive attribute - do not turn attribute off

This option is very handy when backing up files. If this option is specified, only files with the archive attribute will be compressed, and the archive attribute will be set to off when the ZIP operation is complete for these files.

You may alternatively ZIP all files with the archive attribute set, but not modify the archive attribute.

A detailed explanation of the use of this option is given in the *PKZIP Backup Guide*.

Archive Attribute

Every file on an MS-DOS partition has several properties associated with it. One of these properties is called the Archive Attribute. When a file has just been created, this attribute is set to be ON. Also, if a file is altered, the attribute is set. After a file has been backed-up by a program which uses this attribute, the attribute is switched off. By making use of the archive attribute you may make certain that you get all files that are new or changed. You save time by not backing up files you have previously archived. This process is called an Incremental Backup.

See Also: *PKZIP Backup Guide*,
-f, -u, -a+

 -j<h,r,s> Specify masking of file attributes

-J<h,r,s> Specify retention of file attributes (used to override configuration file)

When PKZIP archives a read-only file (or is forced to archive a hidden or system file through use of the -w option) the default is to retain these attributes with the file in the archive. You may choose to strip these attributes off the file during the ZIP process by specifying the -j (lower case) with the letter(s) corresponding to the attribute you wish to mask.

File Attributes

Every file on an MS-DOS partition has several properties associated with it. These attributes are: archive(**a**), hidden(**h**), read-only(**r**), and system(**s**). The archive attribute is explained under the **-i** option. The Hidden attribute prevents MS-DOS from displaying the file in a directory listing. Files that have this attribute are normally ignored by PKZIP. The read-only attribute prevents a file from being changed or deleted. When a file has this attribute you cannot delete the file or change its contents. The system attribute signifies a file that is a system file. System files are used by MS-DOS and are treated as Hidden files. Please see your DOS manual for additional information.

You may set masking of particular attributes to be the default by using either the PKWARE.INI file or the PKZIP configuration program.

If any of these options are set in the configuration they may be overridden temporarily by use of the **-J** (upper case) option.

```
pkzip markie.zip -Jhs c:\util\*.*
```



Please note there is no space between the **-j/J** option and the **h,r** or **s**.



- k** Retain original .ZIP file date
- k-** Set .ZIP file date to the current date (used to override configuration file)

Normally, when PKZIP updates an existing .ZIP file by either adding to it or deleting from it, the .ZIP file date is changed to the current date. If the **-k** option is specified on the command line the .ZIP file date will not be altered.

This option may be made the default, see *PKZIP Configuration*.

If PKZIP is configured to keep the original date in the configuration file, this may be overridden temporarily by use of the **-k-** option.

See Also: **-o**

-
- l** Display license screen

If this command is specified, PKZIP will display the software license agreement.

-
- m[u|f]** Move files

Normally, when PKZIP archives a file, it is making a copy of that file. The original file is not modified in any way. By choosing the **-m** option, you may actually MOVE the file into the .ZIP file.

After the ZIP operation is complete, PKZIP will delete all the files it just placed into the .ZIP file. In effect moving the files from their previous location into the .ZIP file.

This option may be combined with the Update (-u) and Freshen (-f) options. When this is done PKZIP will perform the Update or Freshen as it would normally, and then delete all the files that were placed into the archive.

See Also: -f, -u



-o Set .ZIP file date to newest file in .ZIP file

-o- Set .ZIP file date to current date (used to override configuration file)

Normal default operation for PKZIP is to set the date of a .ZIP file it is modifying or creating to the current date. By specifying the -o option you can tell PKZIP to set the date of the .ZIP file equal to that of the newest file within the .ZIP file.

This option may be made the default, see *PKZIP Configuration*.

If this option is set, you may override it by using the -o- option. This will force PKZIP to set the .ZIP file date to the current date. Note that the -o- option must be completely separate from any other options, such as the -f in the example:

```
pkzip oldfiles.zip -o- -f c:\wp51\text\*.wp
```

See Also: -k



-P Preserve specified path names

-P- Turn off path storage (used to override configuration file)

This option and the related -rp option control the storage of pathnames with files inside of a .ZIP file. The -p (lower case) option is used only in combination with the -r option as -rp (see -r for more information). The -P (upper case) option can be used alone or as a -rP option.

The -P option causes PKZIP to store pathnames found on the command line as part of a filename specified for compression. This command will only have an effect if a file or files specified either on the command line or in a list file have path names.

```
pkzip pathtest.zip -P c:\dos\attrib.exe
```

In the above command the file "attrib.exe" would be compressed into the .ZIP file "pathtest". Because the -P option was specified, and because the path "\dos" was specified as part of the file name, the

file will be stored as "dos\attrib.exe". This means that the path name will be available for reconstruction during extraction. Stored path names can be seen when the .ZIP file is viewed.

```
pkzip path2.zip -P d:\coreldrw\*.cdr
```

In this example all files with the "cdr" extension will be placed into the .ZIP file "path2", and will have the path "coreldrw\" stored with them.

This option may be made the default, see *PKZIP Configuration*.

If path storage is turned on in the configuration file, you may turn it off for a particular command by using the -P- option.

```
pkzip mydoc.zip -P- c:\wp51\docs\zoofund.wp
```

See Also: -r, -v



-q Enable ANSI comments

-q- Disable ANSI comments (used to override configuration file)

PKZIP retains ANSI control codes in comment text that is added to a .ZIP file. During display of the comment text, PKZIP will normally disable ANSI codes. If you want to have the ANSI control codes used by PKZIP when displaying comment text, you must specify this option.

```
pkzip spacemp.zip -q se.diz
```

This option may be made the default, see *PKZIP Configuration*.

If the ANSI option is turned on in the configuration file you may temporarily override it by using the -q- option. Note that the -q and -q- options must be separated from any other options specified.



-r Recurse subdirectories

-rp Recurse subdirectories, preserve the path structure below

-rP Recurse subdirectories, preserve path structure specified and below

-r- Turn off recursion (used to override configuration file)

When this option is specified, PKZIP will search the source directory for subdirectories. If any subdirectories are found, PKZIP searches them for files. If a further level of subdirectories is found in a subdirectory, PKZIP will search these as well.

PKZIP will enter as many subdirectory levels as exist.

```
pkzip mail.zip -r c:\mymail\*.*
```

Using the above command, all files contained within the “mymail” subdirectory, as well as all files within any subdirectories, would be archived. Only the file names would be retained, and not the names of the subdirectories.

```
pkzip mail2.zip -rp c:\mymail\*.*
```

The second example would preserve the same files as the first, but also keep the names of any subdirectories found below the “mymail” subdirectory. Note that the “mymail” subdirectory itself would not be retained since the -P is not being used.

```
pkzip pay_june.zip -rP d:\payroll\*.*
```

The third example will compress all of the files in the payroll directory, as well as all of the files in any subdirectories. The -rp would cause the subdirectories to be recursed and stored. The -P will cause “\payroll” to be stored as well. Without the -P, the “\payroll” portion of the path would not be stored.

This option may be made the default, see *PKZIP Configuration*.



Please note that paths are not stored when the -r option is specified without the p or P option. In order for paths to be stored, the -r option must be accompanied by the -p or -P option.

-s[password] Encrypt files

By specifying this option you may encrypt data as it is compressed by PKZIP. This feature of PKZIP causes the information contained within a .ZIP file to be scrambled. A file which has been scrambled will be unreadable until extracted with the same encryption key.

If a password is not specified the user will be prompted for one.



This feature should be used with caution. For a complete explanation of the use of this option please see *Using Data Encryption*.

-t<date> Select files NEWER than or EQUAL to date

-T<date> Select files OLDER than date

This option will allow you to select files according to their date. The date specifier is a six-digit number conforming to the following formats:

“American” style: mmddyy
040193 April 1st, 1993
103189 October 31st, 1989

“European” style: ddmmyy
010493 1 April, 1993
311089 31 October, 1989

“Japanese” style: yymmdd
930401 1993 April 01
891031 1989 October 31

PKZIP adapts itself to the date format for which your machine is configured automatically.

All users within the United States use the first date format.

If you are outside the United States and are unsure of the configuration of your computer, enter the DATE command at the DOS prompt. This will show you the date format being used on your computer.

The -t and -T options may be combined to specify files from within a date range.

These examples all use the “American” date format (mmddyy). To compress all files created after or on November 4, 1992:

```
pkzip test.zip -t110492
```

To compress all files created before January 10, 1991:

```
pkzip test.zip -T011091
```

To compress all files created between October 1, 1992 and October 31, 1992:

```
pkzip test.zip -t100192 -T103192
```

If -t is used with no date, files with dates equal to the current date are specified:

```
pkzip today.zip -t
```

-u Update files

This option only affects PKZIP operations involving an already existing .ZIP file. When -u is used in conjunction with a PKZIP command, the files specified for archiving will be compared against the files already present in the .ZIP file. If the file to be added into the .ZIP file is already present and is not NEWER, PKZIP will not bother to re-compress the file.

By using this option, you may save yourself time when archiving files that are backed up repeatedly. This option differs from the behavior of the -f option in that files which are not already present in the .ZIP file will be added.

```
PKZIP p:\backup\sdsources.zip -u d:\c700\work\sd*.*
```

In this example a .ZIP file called “sdsources.zip” is created in a “backup” directory on the P: drive. All files in the source directory matching the file specification will be added or updated.

See Also: -f



-v[b][r,c|d|e|n|o|p|s][m]

View [brief] file information

-vt[m] View Technical file information

This option will display technical information concerning the files contained within a .ZIP file. There are several options available with the View option. Each of the -v options may have an ‘m’ added to it, causing the screen to pause every 25 lines.

In it’s simplest form the -v by itself gives a display similar to this:

```
pkzip am70 -v

PKZIP (R) FAST! Create/Update Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745

■ 80486 CPU detected.
■ XMS version 3.00 detected.
■ Novell Netware version 3.11 detected.

Searching ZIP: AM70.ZIP - PKWARE Support BBS (414)354-8670

Length  Method      Size  Ratio   Date      Time      CRC-32  Attr  Name
-----  -
76812  DeflatX    23139  70%    01-30-92  12:03    971fea04  --w-  AM.DOC
191872  DeflatX    80773  58%    01-31-92  15:02    25cf904a  --w-  AM.EXE
11664   DeflatX     6935  41%    01-30-92  11:57    183b5606  --w-  AM.HLP
149567  DeflatX    63211  58%    01-31-92  15:02    bf2fffee  --w-  AM.OVR
2262    DeflatX     1021  55%    03-16-91  10:39    50b63fc5  --w-  NEW-WARE.BBS
5327    DeflatX     2068  62%    10-14-91  11:25    924efc72  --w-  PATHINFO.TXT
130     DeflatX      118  10%    11-14-90  10:05    fd1b1183  --w-  PROGMENU.AM
3664    DeflatX     1644  56%    04-04-91  10:32    c8d5c9cf  --w-  PROGMENU.DOC
1688    DeflatX      740  57%    11-10-90  11:27    8ef23b8a  --w-  READ!.POP
1416    DeflatX      595  58%    01-30-92  12:03    32b1497e  --w-  VENDOR.DOC
2167    DeflatX      949  57%    01-30-92  12:07    2e13f4d4  --w-  WARNING.!!!
2200    DeflatX     1049  53%    01-30-92  12:01    20c953ee  --w-  WHAT'S.NEW
-----  -
448769                182242  60%                                12
```

This is the standard, unsorted view of the .ZIP file contents. The columns displayed from left to right consist of:

<i>Length</i>	Original (non-compressed) size of the file
<i>Method</i>	Type of compression routine used on this file
<i>Size</i>	Current (compressed) size of the file
<i>Ratio</i>	Percent reduction in file size
<i>Date</i>	Date of the file (as it would appear in a directory listing)
<i>Time</i>	Time of the file
<i>CRC-32</i>	The value of the file as calculated by the CRC-32 routine
<i>Attribute</i>	The stored attributes of the file, (system, hidden, writable, read-only, *=encrypted)
<i>Name</i>	Name of the file

The -vb (brief) option will display an identical listing, minus the CRC-32 and Attribute columns.

The -vt option shows additional technical information about a file. When this option is active, a report similar to the following will be displayed:

```
C:\>pkzip am70 -vt am.doc
```

```
PKZIP (R) FAST! Create/Update Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

```
■ 80486 CPU detected.
■ XMS version 3.00 detected.
■ Novell Netware version 3.11 detected.
```

```
Searching ZIP: AM70.ZIP - PKWARE Support BBS (414)354-8670
```

```
      Filename: AM.DOC
      File type: text
      Attributes: --w-
      Date and Time: Jan 30,1992 12:03:54
      Compression Method: Deflate
      Compressed Size: 23139
      Uncompressed Size: 76812
      32 bit CRC value: 971fea04
      Created by: PKZIP: 2.0 under MS-DOS
      Needed to extract: PKUNZIP: 2.0
```

The “File type” is a judgement made by PKZIP during compression based upon the contents of the file. It will be either “BINARY” or “TEXT”

The “Created By” lists both the version level of PKZIP and the operating system used when the .ZIP file was created. Below that is listed the minimum version of PKUNZIP that would be required to extract the file.

Sort Order

By adding one of the following letters to the -v, -vb, or -vt command you can specify the sort order of the display.

- d File Date
- e File Extension
- n File Name
- o Natural Order (Sort order can be permanently specified in the configuration file, the “o” option will override this.)
- p Percentage of compression, smallest to largest
- s File Size, smallest to largest

Any of the above sort orders may be reversed with an “r”.

- r Reverse display order

If you wish to also view the individual file comments with the listing specify “c” on the command line.

```
pkzip hildie.zip -vbcnr
```

This example would display a brief list of the file contents, including individual file contents, and sorted in reverse name order.

View With MORE

```
pkzip totoro.zip -vtrm
```

This example makes use of the “more” feature available with the view command. This causes the display to pause after every 25 lines of information. Adding an ‘m’ to any -v command will cause it to pause every screenful.



-w<h,s>Take hidden, system files

-W<h,s>Skip hidden, system files (used to override configuration file)

Normally PKZIP will skip any Hidden or System files that fall within the files specified for compression. (See the -j option and your DOS manual for an explanation of what Hidden and System files are).

If you wish to force PKZIP to take Hidden or System files you may do so by specifying the -w option, combined with the appropriate

attribute. These attributes will normally be stored as part of the .ZIP file information unless specified with the -J option.

```
pkzip all.zip -whs a:\*.*
```

This example would zip all files in the root of the A: floppy drive, including any hidden or system files.

```
pkzip att_test -whs -Jhs c:\*.*
```

This would take all files as above, and would also discard the file attributes.

This option may be made the default. See *PKZIP Configuration*.

See Also: -j, PKUNZIP -j

-x<filename> Exclude files

-x@listfile.lst Exclude a list of files

Using -x allows you to specify files that you do NOT want used as part of a PKZIP operation. Either a specific filename or a file pattern may be specified. You may use -x multiple times on the same command line, and you may exclude a list of files. The list file is structured and treated identically to all other List Files.

```
pkzip testfile.zip *.txt -xpizza.txt
```

This example would compress all the files with a "txt" extension that are in the current directory, but would NOT compress the file "pizza.txt"

```
pkzip test2.zip *.* -x*.txt
```

Here we take all of the files EXCEPT the files with a "txt" extension.

```
pkzip test3.zip *.* -xcommand.com -xautoexec.* -xconfig.*
```

In this last example all files are specified for extraction, one specific file is being excluded, and any files which match two file patterns are excluded.

See Also: -@ page 46, "Using List Files" page 63

-z Create or modify a .ZIP file comment

Each .ZIP file may have a comment embedded into it. This comment will be displayed when the file is extracted, viewed or modified. A comment may be added to an existing file, or added when a .ZIP file is first created. PKZIP will prompt you for the .ZIP file comment, and allow you to enter up to 127 characters.

It is possible to embed larger comments (up to 4096 characters) by using the following procedure:

1) Create the .ZIP file.

2) Create the comment file (comment.txt). This can be done with any editor that is able to output a standard DOS text file. It must be plain ASCII or ANSI text (if it is ANSI text you must also use the -q option). Although the filename comment.txt is used here, any filename may be used.

3) Issue the command:

```
pkzip zipfile.zip -z < comment.txt
```

This makes use of the MS-DOS redirection ability, redirecting the contents of the file COMMENT.TXT as the input for the comment of ZIPFILE.ZIP.

3a) To add a comment to multiple .ZIP files (within the current directory)

Issue the command:

```
for %f in (*.ZIP) do pkzip %f -z < comment.txt
```

Note that the text of a comment is not compressed. If you have a large message you wish to attach to the .ZIP file it is better to have a short summary and put any longer message in the .ZIP file itself.

Technically, a .ZIP file comment may be as large as 64k, but PKUNZIP will not display a .ZIP file comment this large.

-! **Encrypt authenticity verifiable data into the .ZIP file**

Using this command option will cause PKZIP to embed Authenticity Verification (AV) information within the .ZIP file it is creating.

This option may only be used if you have received and installed the AV information for your fully registered copy of PKZIP from PKWARE Inc.

Note that AV codes for PKZIP 2.0 and above are different than those used for version 1.10.

Please see the section on *Authenticity Verification* for complete information.

@[filename] Specify LIST file

This option specifies a plain ASCII text file to be used as the list of files to be archived. PKZIP will read in this file listing and generate the complete list of files that must be processed. A list file may contain wild card specifications (?,*) as well as exact file names:

```
files.bbs
*.txt
*. *
text\news.asc
d:\text\*.ans
c:/dos/edit.com
```

All of the above would be valid entries in a list file. Notice that both forward and backward slashes may be used and mixed.

The size of a list file that may be processed is limited. The total size of the list file depends upon two factors.

The first is path storage. If paths are specified in the list file and are NOT stored with the -P option, the list file size is limited to approximately 7000 bytes.

If paths are specified and are stored, the list file size is limited to approximately 9700 bytes.

If a list file has drive letters specified as part of the file names within the file, the maximum list file size is reduced. When paths are not stored the maximum size is 5100 bytes, when they are stored 6600 bytes.

List File Size Limits	No Path Storage	Path Storage(-P)
Drive Letters in List File	5100 bytes*	6600
No Drive Letters	7000	9700

*These numbers are approximate and may vary depending on your system configuration.

If the list file is too big, you will receive an “Insufficient Memory” error. If you have a list file which is too big, you may split it into two or more smaller list files and perform multiple PKZIP operations.

See Also: “Using List Files”, -@

-& Create multi-volume archive

When this option is specified PKZIP will create a .ZIP file which is larger than a single floppy disk. This option may only be specified when creating a .ZIP file on removable media (i.e. a floppy disk or high capacity cartridge drive).

If PKZIP encounters an unformatted floppy disk it will format it before it continues. This means that you will not need to anticipate the exact number of floppies that will be needed ahead of time.

For complete information on the use of this and related options, please see the *PKZIP Backup Guide*.

-\$[d] Store Volume label as part of .ZIP archive

When this option is specified PKZIP will store the Volume Label of the specified device as part of the information within the .ZIP file it is creating or modifying. The letter of the drive containing the desired label should be specified in place of the [d] on the option statement. Do not include a colon.

```
PKZIP dougie.zip -rp -$b b:*. *
```

In this example a .ZIP file called “dougie” is created in the current directory. It consists of the complete contents of the B: drive including it's subdirectories and Volume Label.

Volume Label

Every drive that can be accessed on your computer is referred to as a VOLUME. This includes not only your hard disk but also floppy disks, and may include other storage devices such as CD ROM drives or large capacity removable drives such as those manufactured by SyQuest or Iomega. Each of these drives may have a VOLUME LABEL. This is a text string which identifies it. This is particularly useful on removable media, such as floppy disks. Each floppy may have a label which is meaningful to you, or to a piece of software. For further information on VOLUME LABELS please refer to your DOS manual.

-@filename.lst Generate LIST file



Please note that this option is different from using a list file. When PKZIP is reading a list file there is no dash in front of the @.

This option causes PKZIP to generate a LIST file out the specified file name instead of performing a ZIP operation. The output file will be a plain ASCII file with one file name per line, and will be identical to the listing of the files PKZIP would have placed into a .ZIP file had the -@ option not been used.

If the -rp is specified the list file will include paths in addition to the file names. If the pattern/filespec is on a different drive, the drive letter will be included as well.

```
pkzip dummy.zip -@test.lst *.* *
```

This would generate a list file called “test.lst”. Note that the dummy zipfile name is needed. The name of the .ZIP file specified does not matter as it is not actually used.

```
pkzip dummy.zip -@test2.lst -rp d:\*.*
```

This would create a list file of the ENTIRE contents of the D drive. If the current directory is not D:\ when this command is issued the drive will be included with each filename in the list file.

This option can also be useful in conjunction with an existing .ZIP file. For example, to generate a list of all files which have been modified since they were placed in the .ZIP file "akiko.zip":

```
pkzip akiko.zip -f -@modified.lst *.*
```

See Also: "Using List Files"

-= Open file in compatibility mode

This option will cause PKZIP to open files in "compatibility" mode, as opposed to "sharing" mode.

Using this option will cause PKZIP to ignore most file locking, including share.

In some networking situations this will allow you to read a file which would otherwise not be accessible due to file locking.

PKTMP

During compression, PKZIP uses a temporary file. As each file is added to the .ZIP file it is first compressed into this temporary file and then appended to the .ZIP file.

By default this temporary file is created in the current directory.

You may have this temporary file created in a different directory by setting the environment variable PKTMP. Set this equal to the drive you wish to use as your temporary file location.

For optimal performance on your local harddrive, the PKTMP could be set equal to a RAM drive if a large enough one is available. The temporary file location must have enough free space to hold the largest file in it's compressed state.

For optimal performance when creating a .ZIP file on a Novell Netware volume, set the PKTMP equal to any drive on the same server. This will achieve the best performance, and is superior to using the local hard drive or the RAM drive.

```
C:\>set pktmp=z:\
```

PKUNZIP

Command Reference

Description of command options

Items contained in brackets are optional and need not be included. Where there is more than one item separated by commas you may use any ONE of those options following the main option character. Where there are options in brackets you must include that option, or choose one of those in the series. When choosing options from a bracketed series, you may use more than one of these options if they are separated by commas. If the options are separated by a vertical bar (|) then you may only choose one.

Command option Guide

-c Extract files to the console

-cm Extract files to the console with more (pause between screens)

This command will extract files from a .ZIP file and display them on the monitor. The extracted files will scroll continuously on the screen. The software can be told to pause after each screen of output by using the -cm option.

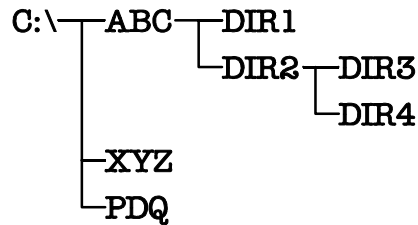
```
pkunzip pkz200.exe -cm pkzip.doc
```

This example lists the text file pkzip.doc to the screen from the self-extracting .ZIP file "pkz200.exe". The display will pause and wait for a key press between each screen of text.

-d Restore directory structure

This command option uses the path names that have been stored in the .ZIP file by PKZIP. If the stored paths do not exist at the extraction destination, they will be created.

For example: If the following directory tree exists on drive C:



If you executed:

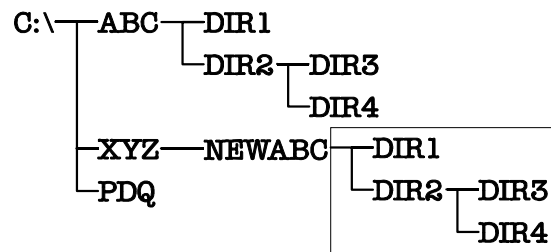
```
pkzip -rp a:stuff c:\abc\*.*
```

A .ZIP file called STUFF would be created containing all the files in the directory “abc”, as well as any subdirectories below that directory.

If you then executed:

```
pkunzip -d a:stuff c:\xyz\newabc
```

The directory structure of drive C: would now look like this:



All of the files and subdirectories which were originally in the “abc” directory are now located in the “newabc” directory (shown in the box). Note that the “newabc” directory was created by PKUNZIP because it was specified on the command line.

See Also: Tutorial - “Storing
and Rebuilding Directory
Structures”

-e[r][c,d,e,n,p,s] Extract files in a sorted order

This option is used to specify a particular order in which to extract files from a .ZIP file. These options are used along with the -e. Each option will cause extraction sorted in ascending order of the particular criteria.

Sort Options

- r Reverse sort order
- c CRC Value
- d Date
- e File Extension
- n File Name
- p Percentage (ratio) of compression
- s File Size

The most useful of these options will normally be extraction in order of Date, Name, or Extension. Extraction in value of CRC order is in effect a random extraction.

In this example the files contained within the .ZIP file ZUCKER would be extracted in the order of their names.

```
pkunzip -en zucker.zip
```

To reverse the order of extraction combine with the [r].

```
pkunzip -ern zucker.zip
```

-f Freshen files in extract directory

This function is the inverse of the freshen function in PKZIP. When this option is used, the PKUNZIP command will be executed normally except that files will be extracted only if:

- The files *already exist* in the target directory or directories
- and
- The files in the .ZIP file are *newer* than the files in the extract directory.

-h Display command summary (Help)

This option will display a brief summary of the command switches available.

-j<h,r,s> Specifies masking of file attributes (default)

-J<h,r,s> Specifies extraction of file attributes

During normal use, PKUNZIP does not extract files as being hidden, system or read-only. If there are files stored within a .ZIP file that have any of these attributes, you must specify the -J (upper case) option to keep the attributes.

```
pkunzip stuff.zip -Jhsr
```

In this example all the attributes would be preserved. For an explanation of what file attributes are see your DOS manual.

See Also: PKZIP -w, PKZIP -j

-l Display license information

This command will display the software license agreement.

-n Extract only newer files

This option is the complement of the -u option in PKZIP. When this option is used files will only be extracted if:

- They do not exist already in the target directory

or

- They are newer than the file already in the target directory

This means that files which already exist and are more recent than those in the .ZIP file will not be overwritten.

-o Overwrite previously existing files

-o- Never overwrite files

This option will cause PKUNZIP to overwrite any files that already exist in the target directory, without prompting. When this option is not specified PKUNZIP will pause if it attempts to extract a file that already exists. It will prompt you:

```
PKUNZIP: Warning! File XXXXXX already exists. Overwrite (y/n/a/r)?
```

If you forget to specify the -o when you give the PKUNZIP command, you may respond to the above question with "a" (All). PKUNZIP will continue as though a -o had been specified originally.

The -o- option will cause PKUNZIP to skip any file which already exists, regardless of file sizes or dates.

-p[a/b] [c] [#] Extract files to the printer

This option is used to extract files from a .ZIP file and send them to a printer (serial or parallel) for printing. The command parameters are defined and several examples are given here to make this complex command clear.

-pa Extract to the printer device in ASCII mode

-pb Extract to the printer device in BINARY mode

ASCII mode is commonly used to print text. It will form feed after each file when multiple files are being extracted.

Binary mode does not send any extra characters, extracting purely the contents of the file to the specified device. This is commonly used to send fonts or graphics to a laser printer. It will not form feed between files.

The a and b options cannot be mixed in the same command line.

In combination with the -pa or -pb option you may specify the port to which PKUNZIP should extract. If only a number between 1 and 4 is specified, the extraction will go to that number of printer port (LPT1 through LPT4). If a "c" is specified PKUNZIP will extract to the COM port. The "c" option may also be combined with a number from 1 to 4, specifying COM1 through COM4.

This example will extract the specified files to the parallel port LPT1.

```
pkunzip sesame.zip *.doc -p
```

This example is extracting files in binary mode to a printer on the second serial port.

```
pkunzip graphix.zip *.eps -pbc2
```

Here the files are extracted to a printer on the second parallel port in ASCII mode.

```
pkunzip docs.zip -pa2
```

-q Enable ANSI comments

-q- Ignore ANSI control codes (default)

By default, ANSI control codes embedded in comments are not displayed by PKUNZIP. When the -q option is used these ANSI sequences will be output and interpreted by the ANSI device driver if one is present.

This option can be forced on by default in the configuration file. The configuration option to add is:

ANSI=enabled

The enabling of ANSI control codes in the configuration file may be overridden at the command line by using -q-.

ANSI control codes provide the feature of adding color to text on a system which has ANSI.SYS loaded in the CONFIG.SYS.



However, ANSI control codes also perform many other features, some of which can be potentially hazardous if you do not know the source and contents of the comment. It is for this reason that PKZIP and PKUNZIP filter out ANSI control codes unless you explicitly tell them not to.

PKWARE has created an addition to the ANSI.SYS driver called PKSFANSI (PK Safe ANSI). By using this in conjunction with your normal ANSI.SYS you can be sure of receiving only the ANSI characters that you want. PKSFANSI is provided by PKWARE free of charge. It is included on the PKZIP distribution disk.

See Also: PKZIP -q

-s[password] Decrypt files

This option is used to unscramble (decrypt) the files in a .ZIP file. PKZIP has a -s option which is used to encrypt the files. If a password is not specified the user will be prompted for it.

When extracting a .ZIP file that has been encrypted, the files will not be extracted unless the correct key (password) is included on the command line or entered at the prompt.



Notice there is *no space* between the -s and the encryption key.

```
pkunzip payroll.zip -o -sSecret *.dbf
```

In the above example, all files with a “dbf” extension will be extracted and the key “Secret” applied to them. This will only work if the files have been compressed with the key of “Secret” applied to them:



This option is case sensitive. This means that “Secret”, “secret” and “SECRET” are all different passwords.

See Also: “Using Data Encryption”, PKZIP -s

-t Test .ZIP file integrity

This option is used to test files within a .ZIP file in order to verify that they are valid and have not been corrupted. PKUNZIP will perform all the actions that would normally be taken during a .ZIP file extraction, except that files will not be written to disk.

The data will be extracted and checked against the CRC to ensure it's validity. After each file is checked it's name will be displayed with an "OK".

```
C:\PKSTUFF>pkunzip zc102 -t

Searching ZIP: ZC102.ZIP - PKWARE Support BBS (414)354-8670
Testing: ZC.DOC          OK
Testing: ZC.EXE          OK
Testing: ORDER.FRM       OK
Testing: VER-HIST.TXT    OK
```

In this example a command is issued which performs a test on the .ZIP file "zc102". As each file is tested its name is displayed. Note that the .ZIP file comment is displayed.

If you test a file that is not a .ZIP file, or specify particular files within the .ZIP file that do not exist, you will receive the message:

```
PKUNZIP: (E11) No file(s) found.
```

Testing a .ZIP file with errors would produce a display something like this:

```
Searching ZIP: PROBLEM.ZIP
Testing: WORLD.M~      PKUNZIP: (W15) Warning! file fails CRC check
Testing: CORELDRAW.BPT OK

PROBLEM.ZIP has errors!
```

Normally, a CRC failure as above indicates that a file has been damaged either in storage or transmission. For instructions on attempting to repair a .ZIP file that has been corrupted, see PKZIPFIX in the *Trouble Shooting* section.

See Also: "Trouble Shooting",
 "F.A.Q."

`-v[b][r,c|d|e|n|p|s][m]`

View [brief] file information

`-vt[m]`

View Technical file information

The `-v` option for PKUNZIP is the same as the `-v` option for PKZIP, with one exception. In PKUNZIP the 'c' option for sorting refers to the CRC value. For PKUNZIP this option refers to the Comment of a file.

For complete information, please see the `-v` option in the *PKZIP Command Reference*.

See Also: PKZIP `-v`

`-x<filespec>` Exclude files from extraction

`-x@list.lst` Exclude list of files from extraction

By specifying a `-x` as part of the PKUNZIP command, you may prevent a file or group of files from being extracted. Using `-x` with a list file will allow you to exclude a list of files from extraction.

```
pkunzip fd201.zip -x*.doc
```

In this example all files are extracted from the .ZIP file "fd201.zip" except for those with a ".doc" extension.

`-x` may be used more than once on the command line, and may be used in combination with specific filespecs and list files.

See Also: PKZIP `-x`

`-$` Restore volume label

This command is used to restore any volume label that has been stored in the .ZIP file. It is important to remember that the PKZIP command used to store a volume label *does not store a drive letter* along with the label. Upon extraction, the volume label will be assigned to the target drive.

```
C:\>pkunzip davebk.zip -$ a:
```

This example will extract the contents of "davebk.zip" to the A: drive. If a volume label was stored as part of the .ZIP file, the volume label for the floppy in drive A: will be reset to the one contained in davebk.zip.

See Also: PKZIP `-$`

-@filename.lst Generate LIST file

Please note that this option is different from using a list file. When a list file is to be read by PKZIP there is no dash in front of the @.

Using this option will cause a list file to be generated using the specified file name instead of a normal PKUNZIP operation being performed. The output file will be a plain ASCII file with one file name per line, and will be identical to the listing of files PKUNZIP would have extracted had the -@ option not been used.

```
pkunzip underdem.zip -@showme.lst
```

This would create a list file called "showme.lst" containing a list of all the files present within the .ZIP file. File specs may be used in conjunction with this type of command, in the same manner as a normal extraction command. Also the -e option may be used for sorting. The following example creates an alphabetized list file of only the ".doc" files within it:

```
pkunzip manuals.zip -en *.doc -@docs.lst
```

The extension of ".lst" on a list file is not mandatory, but recommended for clarity.

See Also: "Using List Files",
PKZIP @, -@

PKSFX/ZIP2EXE

The PKZIP utilities offer a self-extraction capability. This means that you can make a .ZIP file into an .EXE file. This file will in effect be able to extract itself.

A PKSFX (PK-Self Extracting) file consists of a .ZIP file which has been transformed into a self-extracting file by the utility ZIP2EXE. There is no utility or file called "SFX" or "PKSFX", but the self-extracting file will show PKSFX on the banner in the same manner as PKUNZIP:

```
PKSFX (R)  FAST!  Self Extract Utility  Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKSFX/h for help
PKSFX Reg. U.S. Pat. and Tm. Off.  Patent No. 5,051,745
```

Creating a Self-Extracting File

In order to create a self-extracting file, you must first create a .ZIP file. Once this .ZIP file is created, use the ZIP2EXE utility to transform this .ZIP file into an .EXE file.

```
D:\>zip2exe test.zip

ZIP2EXE (tm)      Self-Extract Creator      Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. ZIP2EXE/h for help

TEST.ZIP => TEST.EXE

D:\>
```

As you can see, an .EXE file by the same name as the .ZIP file is created. The original .ZIP file is not deleted.

Meet Junior



Turning a .ZIP file into an .EXE file increases the size because of the included extraction program. The added size is approximately 13.5K bytes. If space is at a premium, and you are willing to sacrifice functionality for size, try PKSFX Junior™.

The PKSFX Junior is somewhat smaller, and offers very few command line options due to this reduced size. The options which are available with the PKSFX Junior are indicated on the PKSFX command summary.

To create a PKSFX Junior file, add the command line option -j when performing the ZIP2EXE step.

```
D:\>zip2exe -j test.zip
```



PKSFX Junior adds approximately 3K bytes to the .ZIP file.

PKSFX Junior can handle a maximum of 512 files. Although PKSFX Junior has this limit to the number of files it can handle, there is no limit to the file size.

PKUNZIP Junior

PKUNZJR.COM

The PKUNZIP Junior program is an extremely small stand-alone extraction program. It is used in the same manner as PKUNZIP but it has only the functionality of the PKSFX Junior self-extractor.

```
PKUNZIP(R) FAST! Mini Extract Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved.
PKUNZIP Reg. U.S. Pat. and Tm. Off.
```

Usage: pkunzjr [-o] filename[.zip] [output_path]

PKUNZIP Junior has one option, -o. This option is the same as that for PKUNZIP (Senior).

PKUNZIP Junior accepts the .ZIP file name on the command line, and may optionally accept an output path. PKUNZIP Junior cannot extract specific files, but must extract the entire .ZIP file.

PKUNZIP Junior cannot extract a file containing more than 512 files, and does not support multiple volume .ZIP files.

PKSFX Command Summary

Command line syntax and options are the same as PKUNZIP. The options listed below are offered by a self-extracting file.

Syntax:

```
pkssf x [option] [d:\path] [file]* [file]*
```

PKSFX indicates the name of the self-extracting file you are executing. No options are necessary if you wish to simply extract all files to the current directory. If you wish to extract the contents to another directory, indicate the drive and/or path the files should extract to. To extract only particular files, indicate the name(s) at the end of the command line.

Options:

@listfile.lst	Specify list of files for extraction*
-c[m]	Extract to console*
-d	Re-create directories stored in .ZIP file*
-l	Display software license agreement*
-n	Extract only newer files*
-o	Overwrite existing files
-p[a b][c][#]	Extract to printer*
-s<password>	Decrypt with password*
-t	Test file integrity*

*Indicates options not available in the PKSFX Junior and PKUNZIP Junior programs.

Options should be placed after the self-extracting files name. If multiple options are used they must be separated by a space and each must have its own option indicator (either - or /).

All options behave in the same manner as they do with PKUNZIP, with one exception. The -s option cannot accept keyboard entry. A password entered with a self-extracting file must be able to be entered from the command line.

Modifying a Self-Extracting File

A self-extracting file can be operated upon by PKZIP and PKUNZIP in a manner identical to normal .ZIP files. You will need to specify the full file name including the EXE extension.

```
C:\>pkzip test.exe -f d:\*.*
```

There is no difference between operating on a self-extracting file and a normal .ZIP file.

A Note About Self-Extracting Files

When you create a self-extracting file, you are adding a copy of the extraction code to that file. Since this code duplicates the functionality of the PKUNZIP program, it makes sense to use the self-extractor only when PKUNZIP will not be present on the machine to extract the file.

If you create many self-extracting files for your own use, you will be using more space unnecessarily. Therefore it is recommended that you only use the self-extracting capability when you are sending a file to someone that you know does not have a compatible PKUNZIP.

Stripping the PKSFX Code

You may have occasion to want to keep the contents of a self-extracting file, but wish to turn it back to a normal .ZIP file.

The ZIP2EXE offers an option to reverse the process. This is done with the command line switch -e:

```
F:\>zip2exe -e test.exe
```

ZIP2EXE will recognize normal and PKSFX Junior files automatically.

Sending a Self-Extractor

If you are sending a PKSFX file to someone who is not familiar with data compression and/or PKWARE products, we recommend you tell them to do the following:

With PKSFX Junior:

- 1) Create a temporary directory on the hard drive to hold the files contained in the self-extracting file.

```
C:\>md temp
```

- 2) Change to the directory that was just created.

```
C:\>cd temp  
C:\temp>
```

- 3) Place the diskette containing the self-extracting file into the A: or B: drive.

- 4) Type the name of the self-extractor, including the path to the floppy drive.

```
C:\temp>a:test
```

With normal PKSFX:

- 1) Place the diskette containing the self-extracting file into the A: or B: drive and change to that drive.

```
C:\>a:
```

- 2) Type in the following;

```
A:\>test -d c:\temp
```

You may also wish to create a batch file or text file on the floppy disk containing the above steps.

The contents of the self-extractor will be placed into this temporary directory. They can then examine or move the files at their leisure. The -d option ensures that any subdirectory information included will be re-created, and causes PKSFX to create the temporary directory for the user.

A site license is required by a corporation, company, educational institution, or governmental agency for use of PKWARE software on the total number of computers that will use the software.

Use of PKWARE programs for personal use is covered by your purchase of a single license for PKZIP. A Distribution License is required by a corporation, company, educational institution or governmental agency to use PKWARE programs to facilitate the distribution of software to outside parties.

Please note that if a self-extracting file is used, a Distribution License is required, because PKWARE code is distributed in the form of the PKSFX program combined with the file(s) being distributed.

Please contact PKWARE for more information if you intend to distribute self-extracting files.

Confirm before extract

If a PKSFX file has a .ZIP file comment attached, this comment is displayed before the file is extracted. By imbedding a Control-S character (Hex 13, ASCII 19), you can cause PKSFX to pause.

The display will look like this:

```
Searching EXE: XXXXX.EXE -
.
.
.
(ZIP file comment text)
.
.
.
Do you want to extract these files now (y/n)?
```

Press ☐Y to extract the files, ☐N to abort. Note that the Control-S character is not displayed. This control character may be located anywhere within the comment.

If the user aborts by pressing 'N', an exit code of 12 will be returned by PKSFX. PKSFX Junior will return an exit code of 8.

See: PKZIP -z for more
information on .ZIP file
comments, Appendix D for
more information on exit codes.

Memory Requirements

The memory requirements for PKSFX are similar to those of PKUNZIP.

PKSFX Junior, and PKUNZIP Junior require a minimum of 50K RAM free, and a maximum of 64K RAM free.

Using List Files

PKZIP, PKUNZIP and PKSFX offer a powerful feature; *list files*. List files are simply listings of files. They are used to simplify command lines, and make repetitive tasks less troublesome. List files can allow you to perform some complex tasks which otherwise would not be possible.

List files can not only be read, but can also be created by PKZIP and PKUNZIP. Also, list files can be created and used by other programs, such as StupenDOS™, PKZOOM™ and PKZMENU™.

PKZIP and PKUNZIP use List Files in two ways: files to be included and files to be excluded. If a List File is being created, the contents will reflect a listing of those files which would have been added, deleted, extracted or listed during a normal PKZIP/PKUNZIP operation.

List File text may contain:

- Comments
- File Names
- Wild Cards
- Exact Path Names and Files
- Drive Letters

List File text consists of plain ASCII. These files may be created using any text editor which can produce plain ASCII files, sometimes referred to as “DOS Text”.

All of the following lines would be valid in a List File:

```
clowns.bmp
*.exe
project.*
MSLR?????.QWK
;This is a comment
emp\census.txt ;This is also a valid comment
D:\data\february.wk1
D:\games\majong\*. *
D:\QWK\*.REP
```

A line in a list file is treated in an identical manner to files specified on the command line. Note that only files and comments are valid in a list file. Both forward and backward slashes may be used in a list file.

Reading List Files

A list file can be used in two different ways with PKZIP and PKUNZIP:

- Files to be acted upon in a ZIP operation.
- Files to be excluded from a ZIP operation.

List files are usually specified with an @ (ASCII 64, normally above the 2 on your keyboard). The list file character can be changed by modifying the configuration file. Please note that the list file generation option (-@) will also change to the selected list file character. Avoid using a character already used for another option.

The simplest example of using a List File is when a .ZIP file is going to be created containing the files specified.

If a List File called “test.lst” was created containing the following lines:
*.COM
300Z.TXT

And then used in a PKZIP command:

```
C:\>pkzip sample.zip @test.lst
```

All .COM files in the current directory would be added to the .ZIP file “SAMPLE.ZIP”, as well as the file “300Z.TXT” if it existed.

The contents of a List File that is specified are included. The contents of a List File specified after a -x are excluded. Multiple List Files to be included and excluded may be specified on the command line in addition to file names being listed alone on the command line.

Following are several example usages of List Files, each with a short description of what action would be caused.

```
C:\>pkzip first.zip @abc.lst -x@xyz.lst
```

All files specified in “abc.lst” would be added to the .ZIP file “first.zip”. Any files that match the file specification in “xyz.lst” would not be added.

```
C:\>pkzip second.zip -d @abc.lst -x*.txt
```

All files listed in “abc.lst” would be deleted from the .ZIP file “first.zip”. Any files matching the “*.txt” specification would not be deleted.

```
C:\>pkzip -vc third.zip @xyz.lst
```

A file listing with comments will be shown for all files in "third.zip" which are listed in "xyz.lst"

```
C:\>pkzip -rp backup.zip *.* -x@noback.lst
```

A .ZIP file containing the entire drive would be created, excluding any files which match those in "noback.lst"



Don't forget to use the @ in front of a list file name. If you do not use the @, the list file itself will be added to the .ZIP file.

Generating List Files

The Generation of list files can have many uses. Some suggested ways to use this feature are:

- Dividing Files to be Zipped into logical groups
- Easy exclusion of files already Zipped from a PKZIP operation
- Complex Batch File operations

Keep in mind that a plain ASCII list file can be fed through other programs, like 'sort', and can be used by some other applications such as StupenDOS or PKZMENU.

The list file generated by PKZIP or PKUNZIP when the -@ option is used contains the names of all the files which would have been acted upon in a normal ZIP operation.

In a simple PKZIP command, the list file will contain a listing of all the files which would have been archived by PKZIP. If the list file is generated with a freshen or update command, the list file will contain those files which would have been added/updated in the .ZIP file.

To generate a list file of your entire C: hard drive:

```
C:\>pkzip dummy -rp -@cdrive.lst
```

Note that the "dummy" .ZIP file name is necessary to have a valid PKZIP command.

The list file generated by PKUNZIP will contain those files which would have been extracted.

Using List files to debug

You may find the generation of list files helpful when you are testing batch files. Let's say you have created a batch file which should compress several files as a backup procedure. Perhaps these files are very large and it takes several minutes to compress each one. This means that if you were to test the batch file simply by running it, it could take quite a while to find a problem and fix it.

By generating a list file of what would have been compressed, you can check very quickly to make certain that the proper files are being selected.

List file size

The size of a list file that may be processed is limited. The total size of the list file depends upon two factors.

The first is path storage. If paths are specified in the list file and are NOT stored with the -P option, the list file size is limited to 7000 bytes.

If paths are specified and ARE stored, the list file size is limited to 9700 bytes.

If a list file has drive letters specified as part of the file names within the file, the maximum list file size is reduced. When paths are not stored the maximum List File size is 5100 bytes. When they are stored the maximum List File size is 6600 bytes.

Note that this table corresponds to the total size of all list files being used in a single command.

List File Size Limits	No Path Storage	Path Storage(-P)
Drive Letters in List File	5100 bytes*	6600
No Drive Letters	7000	9700

*These numbers are approximate and may vary depending on your configuration.

If the list file is too big, you will receive an "Insufficient Memory" error. If you have a list file which is too big, you may split it into two or more smaller list files and perform multiple PKZIP operations.

Be Creative

The list file functions give you a whole new level of power with PKZIP. When defining a problem and searching for a solution, don't forget to consider the power of these features.

Using the list file features can give you access to new abilities PKZIP would not have inherently. Here are some example situations:

Removing Old Files

Suppose a .ZIP file 'work' contains 45 files. The contents of 'work' are extracted to a work directory and several of the files are deleted. How do you delete these files that are no longer necessary from the 'work' archive? Doing it manually would be very tedious.

First create a list of the contents of the directory. By redirecting these file names to a list file you can capture an image of what needs to be kept in the .ZIP file.

```
C:\work>pkzip dummy.zip *.* -@keep.lst
```

The list file 'keep.lst' now contains the files we want preserved in the .ZIP file, files not in this list will be deleted:

```
C:\work>pkzip work.zip -d *.* -x@keep.lst
C:\work>del keep.lst
```

The WORK.ZIP file now contains only those files which are also in the work directory.

Sorting .ZIP Files

Files will be added to a .ZIP file in the order that they are listed in the list file. This can be useful if you want the contents of a .ZIP file to be in a particular order.

After generating a list file, bring it into an editor and modify it to meet your need. To create a .ZIP file that is sorted in alphabetical order, you would use the following series of commands:

```
C:\work>pkzip test.zip *.* -@test.lst
C:\work>sort < test.lst > sort.lst
C:\work>pkzip test.zip @sort.lst
C:\work>del test.lst sort.lst
```

The SORT command used above is a standard DOS program. See your DOS manual for further information on this useful feature.

Using Data Encryption

The PKZIP utilities offer the ability to protect stored data through encryption. You should use PKZIP's encryption ability in place of any other encryption on files you intend to compress.

This encryption ability has been made integral to the function of PKZIP and PKUNZIP to prevent the encryption from interfering with PKZIP's ability to compress. If you feel the need to encrypt your data to protect it, and are intending to compress the data as well, you should then use the encryption of PKZIP. Using the encryption capability of a spreadsheet or database program will prevent PKZIP from compressing the data efficiently.

Encryption is enabled in PKZIP by use of the -s option. The encryption key, sometimes referred to as a "password", follows directly after the -s

```
C:\>pkzip test.zip -ssecret d:\payroll\*.dbf
```

Above, all the "dbf" files from the payroll directory have been compressed into a .ZIP file called "test.zip". During the compression they are also being encrypted based upon the key of "secret".

The encryption scheme used by PKZIP is complex and robust. The encryption key is used to generate a number of binary encryption keys which are in turn used to determine how the compressed data is altered. Neither the original key or the derived keys are stored with the .ZIP file, nor are they retained anywhere else for that matter. This means you must remember the encryption key.

IF YOU LOSE THE ORIGINAL ENCRYPTION KEY YOU WILL NEVER BE ABLE TO EXTRACT THE FILE(S).

It is also not possible to derive the original keys by examining the compressed data, or by analysis of original data versus encrypted data. This means that if you lose the key, there is no way to find it. PKWARE Inc. has no special means or technology for deriving the key or extracting the data without the key.

Also note that a minute change in the key completely alters the encryption scheme. This means that the key must be exact. Any change in spelling, capitalization or punctuation causes a key to be completely different. Be certain that you know what the encryption key is and will not lose, forget, or mistype it.

DO NOT TRUST YOUR MEMORY ALONE. WRITE IT DOWN.
--

Capitalization is also important. “Secret”, “secret” and “SECRET” are all completely different encryption keys.

You should realize that a person who knows you could attempt to extract a file you have encrypted by using words they think you will use or can remember easily. This would be items such as:

- ✓ Your Spouse's name
- ✓ Your Social Security Number
- ✓ Your Address
- ✓ Your Dog's name
- ✓ Etc...

Long Keys

It is possible to have extremely long keys, or keys with spaces in them. The length of the encryption key is limited only by the length of a DOS command line.

To use a key with spaces in it, you must enclose the password in quotation marks:

```
C:\>pkzip test.zip -s"mary had a little lamb" *.dbf
```

Passwords of this sort may be easy to remember, while proving to be very secure. Remember to use phrases that are not obvious to someone who knows you.

Complex Keys

If PKZIP or PKUNZIP is specified with only the -s and no encryption key following, the user is prompted for an encryption key.

```
C:\>zip test -s *.com
```

```
PKZIP (R) FAST! Create/Update Utility Version 2  
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKZIP/h for help  
PKZIP Reg. U.S. Pat. and Tm. Off. Patent No. 5,051,745
```

```
Password ? *****
```

```
■ 80486 CPU detected.  
■ XMS version 2.00 detected.  
■ Novell Netware version 3.11 detected.  
■ Using Super Fast Compression.
```

```
Creating ZIP: TEST.ZIP  
Adding: COMMAND.COM Deflating (36%), done.  
Adding: 4DOS.COM Deflating (34%), done.
```

Asterisks are displayed in place of each character as you enter the encryption key. This feature offers added security in the event someone is watching as you type in the password.

This method of entry may be easier or more comfortable to use. This also allows you to enter characters for the encryption key which cannot normally be entered from the keyboard.

In order to enter one of these characters, hold down the  key and type the ASCII value of the character you wish to enter on the number pad. When you release the  key the character equal to the value typed will be entered into the Password field.

The  key and other keys which could not normally be used since they would interfere with the PKZIP command line can be used when the password is entered in this manner.

Passwords entered in this manner are limited to a maximum length of 64 characters.

Multiple Keys

Passwords can be used not only when creating a new .ZIP file but also when files are being added to an already existing .ZIP file. Let's assume you have a .ZIP file called "keyfun.zip" and it contains two files:

```
june91.wk1  
june91.wp
```

We'll also say that neither of these files are encrypted. Now let's add another file to this .ZIP file, and encrypt it while doing so:

```
C:\>pkzip keyfun.zip -sPiZzA sensitiv.txt
```

This will add the file "sensitiv.txt" with a encryption key of "PiZzA".

Now add another file to the same .ZIP file with a different key:

```
C:\>pkzip keyfun.zip -sDukeNUKE diary.txt
```

At this point you have four files in "keyfun.zip". Two are not encrypted at all, and two are, but with different keys. A view of the file would appear like this:

```
C:\>pkunzip -v keyfun
PKUNZIP (R) FAST! Extract Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off.

■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ XMS version 3.00 detected.

Searching ZIP: KEYFUN.ZIP
```

Length	Method	Size	Ratio	Date	Time	CRC-32	Attr	Name
11664	Deflate	6937	41%	01-30-92	11:57	183b5606	--w-	JUNE91.WK1
76812	Deflate	23404	70%	01-30-92	12:03	971fea04	--w-	JUNE91.WP
5327	Deflate	2082	61%	10-14-91	11:25	924efc72	--w-*	SENSITIV.TXT
3664	Deflate	1654	55%	04-04-91	10:32	c8d5c9cf	--w-*	DIARY.TXT
97467		34077	66%					4

Notice the * next to the Attribute column above. This indicates that a file is encrypted.

Now, if you try to extract the file with no decryption key present you will only be able to extract the two files that are not encrypted:

```
PKUNZIP (R) FAST! Extract Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off.

■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ XMS version 3.00 detected.

Searching ZIP: KEYFUN.ZIP
Inflating: JUNE91.WK1
Inflating: JUNE91.WP
PKUNZIP: (W12) Warning! Skipping encrypted file: SENSITIV.TXT
PKUNZIP: (W12) Warning! Skipping encrypted file: DIARY.TXT
```

Extracting the file with a decryption key will extract any file that the key is valid for, but an error will be reported on any files with a different key.

```

C:\>pkunzip keyfun -sPiZzA -o
PKUNZIP (R) FAST! Extract Utility Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off.

■ 80486 CPU detected.
■ EMS version 4.00 detected.
■ XMS version 3.00 detected.

Searching ZIP: KEYFUN.ZIP
Inflating: JUNE91.WK1
Inflating: JUNE91.WP
Inflating: SENSITIV.TXT
PKUNZIP: (W14) Warning! Incorrect password for file: DIARY.TXT

```

How secure is PKZIP encryption?

There are many factors to consider when using PKZIP encryption. There are also several levels and implications to consider.

The first level of security is physical. Who has access to the .ZIP file? You need to be aware of who might be able to get to sensitive data located on your machine, on your floppies, or on your personal network drive.

The potential hazards of using data encryption cannot be stressed strongly enough. Make certain that you will be able to recall what was used for the encryption key. Also be certain that all people who need to be able to access this data have agreed upon how the encryption will be performed.

It does happen that people encrypt very important data, and then go on “vacation” to a distant land, or meet their untimely demise, without first passing on the encryption key. In these sorts of eventualities the data is gone. So consider well before you encrypt or allow the encryption of information. If this information is so vital to you that you feel you must encrypt it, consider what the loss of it would mean to you.

All forms of encryption, including the one used by PKZIP, are open to “brute force” attacks. This form of attack is simply the trying of many passwords until you find one that works.

In order to help you protect your data from this sort of attack we present figures on how long a brute force attack, using a computer, would take. The scenario we present here assumes that your encrypted .ZIP file is being assaulted by a program which is designed specifically to do this.

An encryption key may contain any valid ASCII character, not just A-Z in upper and lower case and punctuation marks. However, most people will just use the latter. The following table is indexed by the complexity of the password. Across the top is the range of

characters used. The simplest assumes that only lower case letters from a to z were used. The next column assumes that all printable characters were used (a to z in upper and lower case, punctuation, brackets, etc.). The last column assumes a password containing the complete range of ASCII characters.

The vertical index is the length of the password used. This impacts the strength of the password greatly. Think of it as a combination lock. A combination lock with only two numbers would be much easier to break than one with three or four numbers.

We recommend that if you need a truly secure encrypted file, use an encryption key of at least six characters.

The last assumption made is about the speed of the attacking program. For the purposes of this table, we assume that 10,000 possible keys are being attempted per second.

Password "Hacking" Time

Key Length	26 characters (a-z)	96 characters (a-z,A-Z,etc.)	256 characters (All ASCII)
3	2 seconds	1 minute	27 minutes
4	1 minute	2.35 hours	4 days
5	19 minutes	9 days	3 years
6	8.6 hours	2 years	891 years
7	9 days	238 years	2,283 centuries
8	241 days	228 centuries	584,546 centuries
9	17 years	21,945 centuries	149,643,989 centuries
10	447 years	2,106,744 centuries	38,308,861,211 centuries

Choose the complexity that you feel meets your needs, but keep in mind all that has been mentioned about losing and forgetting passwords.

These figures represent the state of technology today. PKWARE Inc. cannot predict future technologies which may allow faster attempts at decryption of a .ZIP file.

Note that the above figures do not include the time needed to actually try all valid passwords. This would increase the time by several hundred percent, dependent upon the length of the file.

PKZIP Configuration

Configuring PKZIP with PKCFG.EXE and the PKWARE.INI file

```
PKCFG 1.0                                Copyright 1992, By PKWARE Inc.

+-----+
| ANSI | DISABLED |
| Zipdate | CURRENT |
| Include | NONE |
| Mask | No Masking |
| View | NATURAL |
| Compress | SPEED |
| Paths | NONE |
| Recurse | OFF |
| Listchar | @ |
| Optionchar | - |
| EMS Memory usage | ENABLED |
| XMS Memory usage | ENABLED |
| Novell Network function usage | ENABLED |
| Use 32 bit instructions | ENABLED |
| Use 32 bit DPMI | ENABLED |
+-----+ Use ENTER to change values -----+

Filename: C:\UTIL\PKZIP.CFG
Allows ANSI comments to be displayed. ANSI comments can remap your keyboard.
F1 Save F5 Reset defaults F10 Exit
```

Sample PKCFG screen

PKZIP can have several of its features configured semi-permanently. These configuration changes are made through a configuration file called PKWARE.INI.

This configuration file can be located in one of three places:

- 1) The current directory
- 2) A directory specified with the environment variable PKWARE.INI
- 3) The directory in which PKZIP.EXE is located

PKZIP will search for the configuration in each location in the order listed above. This means that you may have a default configuration file that is easily overridden by using a configuration file in either of the first two locations.



Any option that is set in the configuration file may be overridden from the command line.

The configuration file may either be created manually with a text editor, or through the use of the included utility, PKCFG.EXE

PKCFG

The PKCFG program will look for an existing configuration file in the same locations and order that PKZIP would. If the PKCFG program finds an existing configuration file it will load it. The existing configuration values will be shown in place of the defaults.

A list of options will be displayed on the screen. A highlighted cursor indicates the option you can modify. Use the  and  keys to move up and down through the list.

Options may be toggled with either the  and  keys or by pressing the  key. Some options have complex possibilities. When one of these is modified, a small dialog box with options to be changed will be displayed.

To reset all displayed options to the default values press the F5 key. In order to save the options you have selected to the configuration file, press the F1 key. Press the F10 key to exit the program. If you exit without saving changes, you will be prompted.

PKWARE.INI

The options available in a configuration file are shown in the table on the next page. The “Default Value” column shows the default value of each option.

The default is the value that would be assumed if the option were not specified in the configuration file.

Configuration File Options

Option	Alternate Value	Default Value
ANSI	enabled	disabled
INCLUDE	hidden/system	(none)
MASK	readonly/hidden/system	(none)
RECURSE	on	off
PATHS	all, recurse	(none)
ZIPDATE	keep, latest	current
COMPRESS	maximal, fast, speed	normal
LISTCHAR	(any)	@
OPTIONCHAR	(any)	-
VIEW (type)	brief	long
VIEW (reverse sort)	reverse	(none)
VIEW (sort order)	size, date, name, ratio, comments, extension	natural
EMS	disable	enable
XMS	disable	enable
NET	disable	enable
DPMI	disable	enable
386	disable	enable

Options separated by a slash / character may be used alone or in combination with others. Options separated by a comma may not be combined with the exception of VIEW. With VIEW one option from each of the three categories may be used in combination.

A configuration file consists of ASCII text. This file may be created with any text editor that can create a plain ASCII or 'DOS Text' file.

Examples

Following are several example configuration files with explanations of what they would accomplish.

```
;Example PKZIP.CFG #1
;Config files may contain comments if they are prefaced by a semi-colon
PATHS=recurse
VIEW=brief/reverse/size
COMPRESS=speed
```

The first example configuration file does the following:

- Turns on the storage of paths that are found during recursion. This would make all commands using the -r option behave as though the -rp option was used.
- Any view (-v) command used without options will be shown in brief format, sorted in reverse order of the file size.
- All commands that do not specify the compression type will use the fastest (-es) method of compression.

```
;Example PKZIP.CFG #2
INCLUDE=hidden/system
RECURSE=on
PATHS=all
VIEW=extension
EMS=off
COMPRESS=maximal
```

The second example configuration file does the following:

- Causes PKZIP to archive hidden and system files by default. Hidden and system files are normally skipped by PKZIP and not visible in directory listings.
- Any command will behave as though the -r option has been used
- All paths will be stored, both those specified on the command line and those found through recursion(-r).
- Any view (-v) command used without options will be shown sorted in order of file extension. Note that the view would be long, the default value. If the view command were specified with -vb using the configuration file, the view would be brief, but the sort order specified in the configuration file would still be in effect unless overridden with a command line option.
- EMS usage is disabled.
- All commands that do not specify the compression type would use the highest compression possible (-ex).

Authenticity Verification

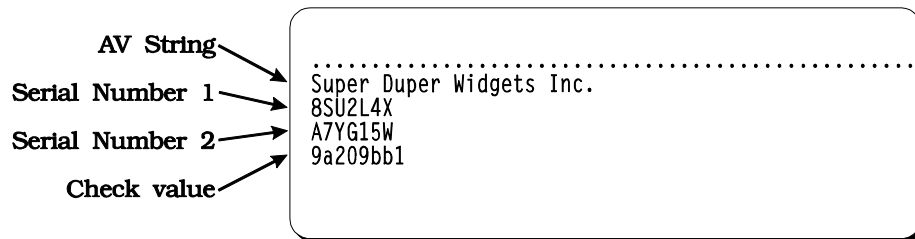
and the PUTAV program

The Authenticity Verification feature allows you to create .ZIP files that PKUNZIP or PKSFX can test for authenticity while extracting. This provides for detection of tampered, hacked, or virus infected files.



To use this feature, you need to be a fully registered user of the software. Mail or fax, to PKWARE, a **photo copy** of the Authenticity Verification application form found at the end of this chapter. PKWARE will then send to you a letter containing the information needed to imbed this AV information into your copy of PKZIP.

The letter you receive from PKWARE will contain your AV information. It will appear as follows:



Using PUTAV

The PUTAV (put-ay-vee) program is used to install the AV information into the PKZIP program. PUTAV.EXE and the PKZIP.EXE files should both be in the same directory.

PUTAV will alter your PKZIP.EXE program. The information you enter will be embedded into the PKZIP.EXE program.

PUTAV Ver 2.00 - Put Authenticity Verification in PKZIP.EXE
Copyright 1990-92 PKWARE, Inc. All rights reserved.

Enter company name exactly as it appears on the PKWARE documentation.
Company Name : **Super Duper Widgets Inc.**
Enter serial numbers exactly as they appear on the PKWARE documentation.
Serial Number 1: **8SU2L4X**
Serial Number 2: **A7YG15W**

Working - done.

Serial number installation complete. Check value: 9a209bb1

After PUTAV prints "Working..." it will display a "twiddle". This spinning graphic is to let you know it is still busy. The PUTAV process may take several minutes.



After PUTAV has finished embedding this information, it will display a "Check value". Compare this number to the one printed on the letter containing the information you entered. If these two numbers do not match exactly, you have entered the information incorrectly and should repeat the process. All AV information is verified at PKWARE before it is sent to you.



A note to PKZIP 1.10 users: The Authenticity Verification feature of PKZIP 2 is completely new. Your old AV numbers will not work with PKZIP 2. PKUNZIP will ignore any AV stamp created by PKZIP 1.10.

How AV Looks

When a file containing AV information is extracted, a success or failure message is displayed.

Success

```
PKUNZIP (R)    FAST!    Extract Utility    Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off.
```

```
Searching ZIP: TEST.ZIP
  Exploding: FILE1.DAT    -AV
  Exploding: FILE2.DAT    -AV
```

```
Authentic files Verified!
# XYZ123    Super Duper Widgets Inc.
```

Failure

```
PKUNZIP (R)    FAST!    Extract Utility    Version 2
Copr. 1989-1992 PKWARE Inc. All Rights Reserved. PKUNZIP/h for help
PKUNZIP Reg. U.S. Pat. and Tm. Off.
```

```
Searching ZIP: TEST.ZIP
  Exploding: FILE1.DAT    -AV
  Exploding: FILE2.DAT    -AV
```

```
PKUNZIP: Warning!
Authenticity Verification Failed!
One or more of these files has most likely been tampered with.
```



The “Authentic files Verified!” message is generated by PKZIP. It is not necessary for you to include any sort of verification message as part of your AV string.

If you wish to include additional information with your AV string, this is easily accomplished. If the file “AVEXTRA.TXT” is present when a .ZIP file is created using AV, the contents are appended after the AV string and will be displayed on the following lines.

This feature is convenient when you wish to include information which may change. In this way you do not need to have your actual AV changed.

Application for Authenticity Verification

In order to receive a serial number to use with the PUTAV program, please fill out and return this form to PKWARE. When the enclosed information is verified, you will receive your serial number from PKWARE.

Please note that this does not constitute a license from PKWARE to distribute the executable programs PKZIP, PKUNZIP or self-extracting PKSFx files for commercial purposes. See LICENSE.DOC or contact PKWARE Inc. for more information about distribution licenses for the software.

Please Read the entire chapter on AV before filling this out.

- 1) My full registration for PKZIP/PKUNZIP/PKSFX is enclosed
(circle one) YES NO

If yes, skip to step 3.

- 2) I have previously fully registered PKZIP/PKUNZIP/PKSFX in the past. The name under which the software is registered is:

Name: _____

Company: _____

Title: _____ Date: _____

Address: _____

City: _____ State: _____

Zip Code: _____

Telephone #: (____)_____ Fax #: (____)_____

Approximate date of registration / purchase: _____

- 3) Please list the name that you want displayed by PKUNZIP when an authentic file created by you is verified (51 characters max).

Since the purpose of the Authenticity Verification feature is to help provide detection of tampered, hacked or virus infected files, we ask that you provide only your company's name or individual's name for the AV string.

The "AVEXTRA.TXT" feature was designed to additionally display telephone numbers, addresses, BBS numbers, BBS nodes or parameter, CompuServe addresses and similar information.

[illegible]

F.A.Q.

(Frequently Asked Questions)

This section lists some common questions about PKZIP and related subjects. We hope you will find this information helpful.

Why do I get the message "Bad Command or Filename"?

If you see this message, DOS is telling you it can't find the program you are referring to. This can happen because you are either not spelling the name of the program properly, or you didn't put a space between the program name and its options, or the program is not in your path.

Please see the *Installation* section and your DOS manual for more information.

Why is this program so complicated?

PKZIP initially was used to compress files that would be sent by modem. The people using PKZIP were mostly hobbyists with a fair amount of computer experience. PKZIP was designed to be powerful and fast.

Someone who is familiar with the workings of PKZIP can be very efficient with the command line. The command line interface is vital for automated or batch use of PKZIP.

Although PKZIP may seem very complex at first, it is easy to master if you take one step at a time. Start simply by following the Tutorial, then experiment and build on that knowledge as you need more features.

Users who prefer using Graphical User Interfaces may wish to consider the "PKZMENU" product from PKWARE. This easy to use menu-driven program offers PKZIP compatibility. It features drop-down menus and a simple point and shoot interface.

Why didn't the files I Zipped get any smaller?

On occasion, you may find that the files you add to a .ZIP file do not compress. These files will be “stored”. This will happen when a file is either already compressed or encrypted.

In both cases the redundancy of the file will be gone, making compression difficult if not impossible. If the file is encrypted you should first save it without encryption, and then use the encryption feature of PKZIP.

You will often find that files distributed with commercial applications are already compressed. A common example would be the graphic data files that come with games like Leisure Suit Larry or Ultima.

I zipped up a bunch of files but now I have LESS disk space!?

When PKZIP compresses files, it makes a copy of the original file. The original file(s) are still present. If you wish to recover space that was taken up by the original file(s), you must either delete them yourself, or instruct PKZIP to delete the file(s) with the -m option.

What's the difference between -u and -f?

The update and freshen options are very similar. This may be confusing at first, but the difference between them is fairly easy to understand.

The freshen option will tell PKZIP to archive any files which match those already in the .ZIP file. These files will only be re-compressed if they are newer than the files already in the .ZIP file. Each file is evaluated individually.

The update option will archive all files, with one distinction. If the -u is not used, all files specified will be compressed and added to the .ZIP file, even if they already exist in the .ZIP file. By using the -u option, you instruct PKZIP to compare what is already in the .ZIP file against what it was asked to compress. If a file is already present in the .ZIP file and is also in the source directory, PKZIP will compress a file only if it is newer than the copy of the file within the .ZIP file. If a file in the source directory is not already present in the target .ZIP file, PKZIP will add it to the .ZIP file.

Is PKZIP compression “lossy” or “lossless”?

PKZIP uses a “lossless” compression scheme. This means that 100% of the original data is preserved and re-created. There is

absolutely no difference between the data that you put in and the data which you get back out.

There are other compression methods that are known as “lossy”. The idea behind these compression methods is that if you throw away some of the data, it becomes less complex and therefore can be compressed more. This type of compression is only useful for data that need not be precise. This applies to some applications that use pictures and sound.

How do I zip up subdirectories?

In order to ZIP up subdirectories you must both *recurse* subdirectories and *preserve* path names. This is done with the -r and -p options in combination. The options may be placed together as -rp.

When a .ZIP file is created with paths stored, these paths will be visible in a *view* of the file (-v).

To re-create these subdirectories, or to place files into their original subdirectories, the -d option must be used with PKUNZIP.

Please see the tutorial for more information.

I zipped up some subdirectories, but I can't get them to come back?

Did you remember to use the -p option in addition to the -r option? You may have stored files that are in subdirectories but not preserved the path names as part of the .ZIP file. To check if there are paths in the .ZIP file, do a view of the file:

pkunzip file.zip -v

If you do not see paths as part of the file names within the .ZIP file, then paths were not stored and cannot be recovered. If you do see paths all you need to do is use the -d (directory creation) option when you extract the files.

How do I unzip a single file that is in a subdirectory in the .ZIP file?

Extracting a particular file from a .ZIP file is simple. You type PKUNZIP with the name of the .ZIP file and the name of the particular file you want. With a .ZIP file that contains paths the procedure is basically the same.

Let's assume a “test” .ZIP file with these files in it:

```
file1.txt
apple\file2.txt
apple\banana\file3.txt
```

To extract only “file2.txt” from this .ZIP file you must specify the complete name. This would include the path.

```
pkunzip test.zip apple\file2.txt
```

Note there is no leading slash. This is how PKUNZIP differentiates between the files you are extracting with paths, and the target directory.

“file2.txt” would be extracted to the current directory. If you wanted to extract it with its subdirectory simply include the -d option on the command line.

How do I unzip a directory without also extracting its subdirectories?

Using the above “test” .ZIP file we could extract the entire contents of the apple subdirectory easily:

```
pkunzip test.zip apple\*. * -d
```

If we did it as shown above we would not only extract all the files in the “apple” subdirectory, but also the “banana” subdirectory below it and any files it contains.

To extract only the “apple” subdirectories contents, and nothing else, we must exclude those directories we do not wish to extract:

```
pkunzip test.zip apple\*. * -d -xapple\banana\*. *
```

If the “apple” subdirectory had multiple subdirectories off it you would need to exclude each one individually on the command line.

What is an AV?

AV is short for “Authenticity Verification”. AV is a process whereby a copy of PKZIP has unique codes and information contained in it identifying the owner of the copy of PKZIP. This information is then encoded into a .ZIP file when it is created.

When the file is extracted by PKUNZIP, this information is checked. If the .ZIP file has been modified by a copy of PKZIP other than the one which initially created it, PKUNZIP will report that the file has been tampered with.

The advantage of this feature is to offer a layer of protection between the creator of an archive and the recipient. The recipient knows that the file received is the file that was sent, as well as being able to identify the creator if it is not known.

Where is my AV number?

As a licensed user of PKZIP you are entitled to one AV number. In order to receive an AV number, you must apply for it.

If you have a need to create AV encoded .ZIP files, fill out the form contained in the file "authveri.frm" found in your PKZIP archive. Mail or fax this form to PKWARE. Please allow up to four weeks for processing time, plus mail delays. The codes needed to install your AV number will be returned to you via US Mail. AV codes will not be returned by FAX, telephone, or any means other than US Mail.

Please note that although the initial AV code is included with your purchase of a PKZIP license, changes or re-issues of an AV code may be at an additional cost.



You must apply for your AV number. The serial number on the disk PKWARE sent to you has no relation to the AV number. The AV number is based solely on the information you provide for your AV identification string on the AV application form.

I forgot my password, what do I do?

1. Read the chapter "Using Data Encryption"
2. Try to remember the password.
3. Try passwords that are "close" to what you think it was.
4. Try mixed upper and lower case versions of your password.



There really is nothing that can be done if you lose or forget your password. Don't forget or lose your passwords! PKWARE has no special means for "getting around" the encryption and may not be able to assist in the recovery of an encrypted file.

YOU SHOULD KEEP A WRITTEN COPY OF YOUR PASSWORD(S) IN A SECURE PLACE. DO NOT TRUST YOUR MEMORY ALONE.

What does "Don't Know How to Handle" mean?

There are many different methods of compression. In the history of PKZIP alone there have been seven different methods to date. The .ZIP file format was designed so that additional methods of compression can be added as they are developed. In this way the .ZIP file format will never need to be abandoned.

If you attempt to extract a .ZIP file that was created with version 2.0 or higher with a lower version of PKUNZIP you will receive the message “Don’t Know How to Handle” for every file compressed with a more advanced algorithm.

If you attempt to extract a .ZIP file created with a version of PKZIP that includes a method PKZIP 2.0 does not use, you will receive a warning that will tell you what version of PKUNZIP you need.

What is a CRC and why did it fail?

CRC is short for Cyclic Redundancy Check. A CRC is a value computed to represent the data content of a file. The CRC is computed by PKZIP when it archives the file, and computed a second time when the file is extracted. If the second CRC does not match the first CRC, then a change has occurred in the data.

The cause of a CRC failure can vary. Here are some possible causes for CRC failures.

- I. Failure on the machine that created the .ZIP file.
 - A. Failure immediately after creation.
 - 1. On a 386/486 or higher there may be an obsolete or poorly written TSR or device driver interfering with PK(UN)ZIP. Use the -3 option with both PKZIP and PKUNZIP to see if this is the problem. Consult the *Trouble Shooting* section for more information.
 - 2. “Clean boot” your machine. Either re-name your autoexec.bat and config.sys files so that they are not loaded, or insert a DOS boot floppy and reset your machine. Attempt extraction and compression of the file again.
 - 3. The floppy/hard disk controller may be failing. See if there is any difference in behavior if you compress to/from the hard disk or floppy drive.
 - 4. Failing diskette media.
 - 5. A bad memory chip.
 - 6. Your computer is suffering from the “cache flaw”. This is caused by problems with the external CPU cache on your computer’s motherboard. See the *Trouble Shooting* section, -(option.
 - B. Failure after a period of time.

1. Any the above causes.
2. The file has been damaged during the time it was stored on the hard disk. The can be caused by cross-linking or "lost" clusters. Some faulty disk-defragmenters and deletion recovery programs may cause this.
3. Deleting and then undeleting a .ZIP file. "Undelete" programs are by no means flawless. An undeleted or recovered file is very likely to fail, as most disk recovery programs are imperfect at best.

II. Failure on a network server.

- A. Some network server software does not run properly on 386/486 machines. If PKZIP is used on the server CPU running this faulty network software, it will be interfered with by the network software.

III. Failure after transfer of a file by modem.

- A. Any of the problems cited in section I may be the cause of a failure.
- B. There was an error in the transfer of the file. Use an "error-correcting" protocol such as ZMODEM to avoid these types of problems.
- C. The file was transferred with "Kermit". If you use Kermit you must set it to Binary mode. PKZIP files are binary.
- D. One of the computers or modems involved in the transfer has a problem.

IV. Failure after transfer by disk.

- A. Any of the problems cited in section I may be the cause of a failure.
- B. Faulty disk controller/floppy drive.
- C. Faulty floppy disk.
 1. The drive alignment between the two computers may be off enough to cause a data error. Normally you will get a "disk failure" message from DOS.
 2. The disk was damaged by a magnetic or heat source.
 3. The disk media is worn out or defective.

What do "Bad Table" and "Inconsistent Local Header" mean?

If you receive either or both of the above messages, the .ZIP file is damaged or has been tampered with. These messages may also be accompanied by a CRC failure.

All of the physical causes listed for CRC failures apply as possible causes of these errors.

How can I make PKZIP run faster?

PKZIP defaults to a compression method that is average in both compression amount and speed. If you want to get the most speed out of PKZIP there are several things you can do:

1. Specify a faster compression method with the -ef or -es option.
2. Use a large RAM disk as your temporary files drive.
PKTMP=(RAM disk drive letter)
3. If you are creating the .ZIP file on a network drive, set your PKTMP equal to a drive on the same server.
4. Make as much conventional memory available as possible.

How many files can be in a .ZIP file?

A .ZIP file may contain a maximum of 16,383 file entries under MS-DOS. Due to memory limitations this limit cannot actually be realized. Under the current memory scheme of DOS the largest number of files which may be in a .ZIP file is approximately 9000 files.

The maximum number of files you can handle with a .ZIP file operation can be calculated by the following formulas:

With No EMS or UMB's available:

$(\text{Free Conventional RAM} - 375\text{k}) / 64 \text{ bytes} = \text{Maximum \# of Files}$

With EMS available:

$(\text{Free Conventional RAM} - 85\text{k}) / 64 \text{ bytes} = \text{Maximum \# of Files}$

The "Super Fast" algorithm uses 32k less RAM than the other methods in PKZIP 2. You will be able to compress an additional 500 files when using the "Super Fast" method.

Note that the value of 64 bytes per file is approximate and may vary.

When a .ZIP file is being updated, you must not only take into account the files being added, but allow 70 bytes per file in the original .ZIP file.

Can I send a .ZIP file to a different type of computer?

As of the publication of this manual, PKWARE currently supports PKZIP on MS-DOS and OS/2 platforms, PKWARE intends to support additional platforms and will announce these as they become available.

Because PKWARE has dedicated the .ZIP file format to the public domain, it is possible for other people to write programs which can read .ZIP files.

NOTE THAT THE PKZIP, PKUNZIP, PKSFX PROGRAMS AND THEIR ASSOCIATED SOURCE CODE AND SUPPORT PROGRAMS ARE THE EXCLUSIVE PROPERTY OF PKWARE INC. AND ARE NOT PUBLIC DOMAIN SOFTWARE PROGRAMS.

We are currently aware of PKUNZIP compatible programs for a number of different platforms. A .ZIP file can be transferred to any platform for which you can find a compatible extraction program.

Extraction and Compression programs not developed by PKWARE may not be completely compatible with the .ZIP file standard.

Contact PKWARE for a list of platforms for which PKZIP and PKZIP compatible software is available.

What is the difference between -b and pktmp?

PKZIP creates two different types of temporary files, and allows you to specify the location for these two types separately. The two situations where PKZIP creates temp files are:

- When a file is being added to a .ZIP file.
- When a .ZIP file is being modified.

It is advantageous to have the temporary files for the first instance written on the fastest drive on your system, unless you are creating a .ZIP file on a Novell Network.

The location for the temporary .ZIP file created when a .ZIP file is modified is normally the same as the location of the original file. In some cases you may not have enough room on that drive. When this occurs use the -b option to relocate the temp file.

How Does PKZIP Work?

This section provides a description of how PKZIP actually does its job. It isn't completely necessary for you to know or understand the information presented here, any more than you need to know how your carburetor works to drive a car. It is presented to help you feel more knowledgeable about the software, for the technically oriented, and for the merely curious.

Two Processes

PKZIP performs two functions; compression and archiving. Although the two ideas may seem related, they are actually completely separate. Compression is the process of representing a given piece of information with a smaller piece of information. Archiving is the process of combining many files into a single unit, usually along with vital information about each file.

Compression

The actual process used by PKZIP for its compression is too complex to explain in detail. Instead some of the general principles behind information theory and compression are explained.

In order to understand data compression, you need to understand two ideas: Information content and Binary coding.

Information content

Everything in your computer, everything you ever read, is "information". The more complex a message is, the higher the information content. The less complex, the less "random" a message is, the lower the information content.

If a message contains a low amount of information, it should be possible to represent it in a smaller amount of space. Look at this page, for example. How much of the page is white space with no letters (information) on it? If you took away all of the white space this page would be about a third smaller. How many times are the words "the", "information" and "compression" on this page? If you could replace each of these words with something smaller you would save a significant amount of space.

The more frequently the same group of symbols (in this case, letters) appears, the lower the information content of the message.

The field of Information Theory uses the term entropy to describe the "true" information content of a message. There are formulas that can be used to determine the entropy of a message. The idea behind data compression is to get a new message representing the old one, with as close to the entropy of the original message as possible.

As a simple example, consider this sentence:

she sells sea shells by the sea shore

This sentence is 37 characters long, including spaces. The spaces cannot be simply thrown away as the meaning of the original message would be lost.

There are fairly obvious patterns to the sentence. The combination 'se' appears three times, 'sh' three times, and 'lls' twice. In fact the 'se' pairs all have a space in front of them, so these can be ' se'.

she sells sea shells by the sea shore

We can replace each of these patterns with a single character:

```
#=' se'  
$='sh'  
%='lls'
```

Note that the first replacement string includes a space at the beginning.

If we reproduce the sentence with these symbols it now looks like:

\$e#%#a \$e% by the#a \$ore

The new representation is 24 characters long, this is a saving of 13 characters, or 36%.

Binary Data Representation

All information used, stored, and processed by computers is represented by two values, zero and one. Everything that you see on your screen, everything stored on disk, is all represented by combinations of zero and one.

You can think of it as a sort of Morse Code. In Morse Code there are also only two values, dot and dash. When a computer stores a character, it uses a combination of eight zeros and ones.

Having eight positions in which to store a zero or one gives the computer 256 different possible combinations. You arrive at this number of combinations in this way:

If you have one coin, it can be in either of two positions:
Heads(0) or Tails(1)

0 or 1

If you have two coins, there are four possible combinations:

00, 11, 10, 01

If you have three coins, there are eight possible combinations:

000, 001, 010, 011, 100, 101, 110, 111

As you can see, each time you add another coin (binary digit) the number of possible combinations doubles: 2, 4, 8, 16, 32, 64, 128, 256.

So the computer uses eight binary digits to get 256 possible values. These values are mapped onto a table called ASCII (American Standard Code for Information Interchange). Each different combination has a particular character that is mapped to it, such as a letter, number or symbol. Each of these positions of 0 or 1 is called a bit.

she sells sea shells by the sea shore

The sample message above would be represented by 296 bits (37x8 bits).

If we follow standard ASCII, we have 256 different symbols being represented for our use. The sample sentence we are using only contains alphabetical characters, and only 11 of them at that. If we only need 11 different values we could use a lower number of bits per character.

The closest value to 11 using binary combinations is 16 combinations, using 4 bits per character. If we wrote a new table of our own using four bits per character, and used it to represent the message, we would use only 98 bits. This would be half as many bits, a considerable savings.

We can do better.

It is possible to have binary codes of varying length. To do this we must use codes with unique values that are not repeated as the beginning of another code. In this way we can find the codes in a long stream of zeros and ones.

If the codes were not constructed to have unique beginnings, it would not be possible to find each individual code within a long stream of zeros and ones.

There are many types of coding techniques which produce codes of varying length, based upon symbol frequency. Some well known coding schemes are Huffman and Shannon-Fanno. PKZIP uses a type of arithmetic encoding. The scheme is too complex to document here, nor is it necessary for you to understand the principles described here.

A table of variable length codes for 11 symbols would look like this:

11	1101
110	0100
101	1000
001	01010
1011	00000
0010	

As you can see, the codes are getting longer and longer. Because of this, we will get the best results if we map the shortest code to the most common symbol in the message. If you know Morse code, or have occasion to look at it, you will notice that frequent characters, such as 'e', 't', 's' and so on have shorter codes assigned to them. Morse code tends to be about 25% more efficient because of this than it would have been had the codes been assigned at random.

A useful idea here is to allow a symbol to be not only a character, but a group of characters.

Using the common patterns found in the first analysis of the message, we can map the following table:

Occurrences	Symbol	New Code	Bits in Message
4	e	11	8
4	(space)	110	12
3	' se'	001	9
3	sh	101	9
2	lls	1011	8
2	a	0010	8
1	b	1101	4
1	y	0100	4
1	t	1000	4
1	o	01010	5
1	r	00000	5

Our new coding scheme can represent the message with only 74 bits. This is a saving of 222 bits from the 296 bits used in the "natural" encoding. This is one quarter of the original message size.

One important factor that would affect a real situation is the table we are using. In order for the data to be re-created from the "compressed" representation, we must include a copy of the table used to encode the data.

This can be a seriously limiting factor. If the data is too complex, or the encoding scheme too inefficient, the table used can be as big as the space saved by the encoding. In the worst cases, an attempt to re-encode the message using a table will result in the encoded message plus the table being larger than the original message!

This is why data which uses a low number of symbols and frequently repeated combinations of symbols, such as a text file, compresses well. Complex, highly random data, such as the information representing a program on disk is difficult to encode efficiently, and therefore compresses less.

Speed vs. Size

Searching for these patterns, and determining an efficient way to encode the data, takes a lot of computer power and time. The more time taken to analyze the data the better the compression will be. To get more speed, you must sacrifice some level of compression.

There are other steps and methods used in powerful compression schemes such as those used by PKWARE products. Hopefully this explanation gives you a better understanding of what happens when PKZIP compresses data.

Archiving

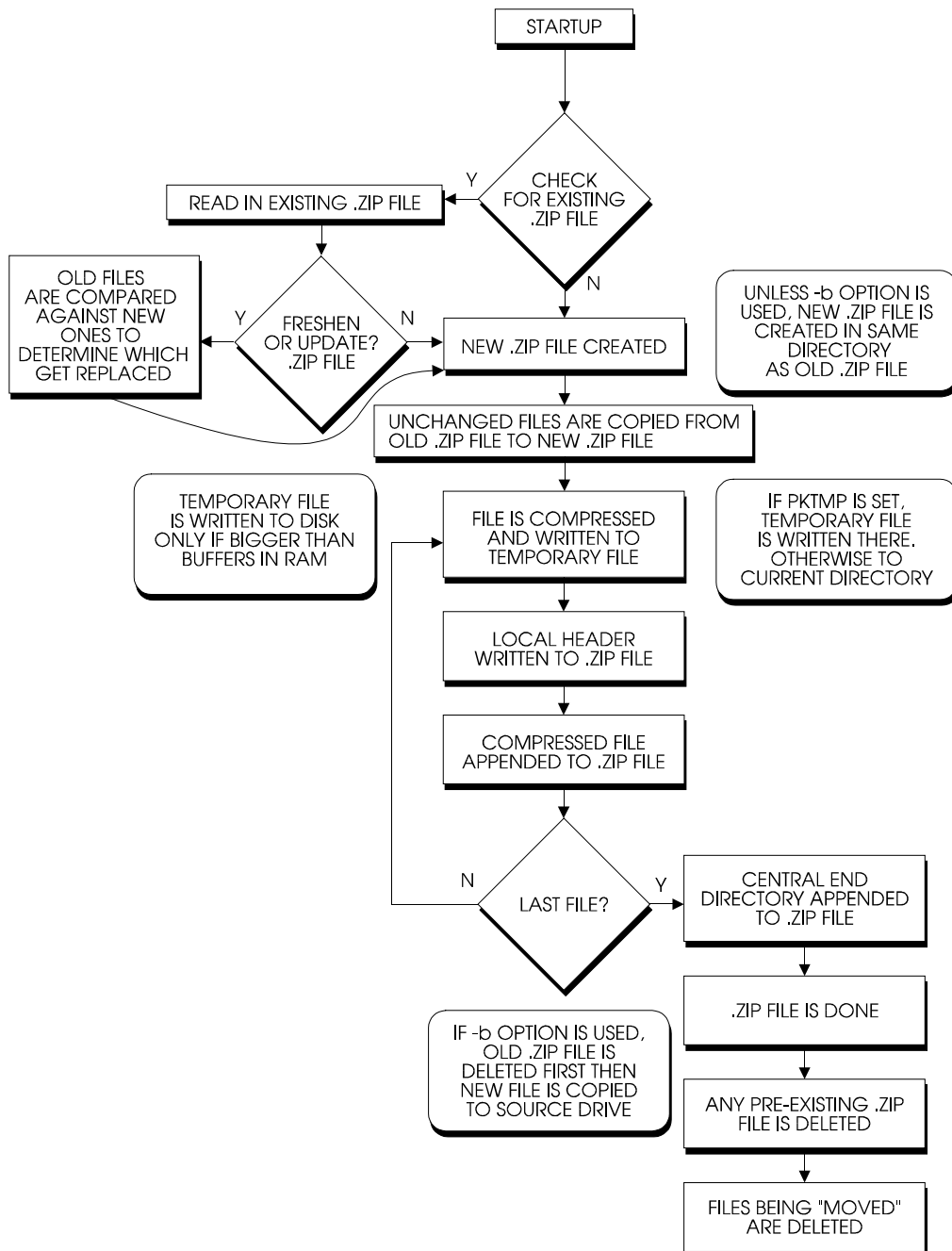
Programs usually rely heavily on associated data files, or may actually consist of several related programs. Some programs may require dozens or even hundreds of files.

In the “dawn” of the PC age, people wanted a way to keep all of these associated files in one spot. “Library” programs were created to take a number of files and group them together into a single file. This made them easier to find, easier to store, and much easier to send to someone by modem. It makes much more sense to be able to send someone a single “package” instead of many files. If you forget a file all sorts of problems arise.

These programs were the birth of archiving. In order for a single file to hold many files, information about each file also had to be stored in the archive. This information could then be used by the archival software to locate a file and pull it out, or to list information about the files contained within an archive.

Compression was first available as a utility that would take a single file, and produce its compressed equivalent. People began to group files together with a Library program and then compress the archive file.

How PKZIP Builds a .ZIP file



The next and obvious step in this process was to combine the two ideas. Compress the files and archive them. This made storage very simple, the compression was no longer a separate step and could be taken for granted as part of the archiving process.

PKZIP 2 is the second generation of these programs. PKZIP can not only compress and archive files, but also stores a great deal of vital information about the files. PKZIP can even store directory structures and volume labels.

How PKZIP Builds a .ZIP File

When you specify a PKZIP command line, PKZIP goes through several steps:

1) Parsing the command line.

2) Determining the exact computer it is running on, looking for:

- The CPU type 8086/8088/286/386/486/etc.
- XMS/EMS memory and UMB's
- A Novell Network

3) It reserves the memory it will need to perform the compression, archiving and buffering.

4) PKZIP then looks for a .ZIP file with the same name as the one you specified on the command line. If it finds one, PKZIP reads the information on the files it contains.

PKZIP then performs the requested action.

Building a New .ZIP File

PKZIP builds a new .ZIP file if it finds nothing in step 4.

5) PKZIP reads the information from the command line specifying what files it is supposed to take, what files it shouldn't take, and if there is an exclude command.

If only a single file is specified, PKZIP looks for this file.

If a group of files is specified, with a wildcard, PKZIP asks DOS for a list of files that matches that pattern.

If a @list file is used, PKZIP reads it, then checks for which files exist. If a pattern is specified in the @list file, PKZIP gets a list from DOS of the files which match this pattern.

If directory recursion has been specified with the -r option, PKZIP next looks for any subdirectories. If it locates subdirectories it goes

into them and looks for any files matching the files specified on the command line or in the @list file. If PKZIP finds subdirectories in the subdirectories it repeats the process. It will continue this process until it finds no additional subdirectories.

Now PKZIP has a list in memory of all the files it should take. The files specified for exclusion are now compared against this list, and any that match are removed. If after this step is complete the list in memory is empty, PKZIP finishes with a message "Nothing to do".

PKZIP can maintain a list in memory of up to approximately 9000 files. If there are too many files in the list it makes in memory, or if you do not have enough memory available, you will get an "Out of Memory" error.

6) Now PKZIP reads in each file, one at a time, and compresses it. When it is finished compressing a file, it adds it to the .ZIP file being created.

PKZIP compresses the entire file first, and then adds it on to the .ZIP file. If there is enough memory available it will keep this compressed temporary copy in memory. If there is not enough memory available it must create a temporary file on disk, and then copy this to the .ZIP file.



This temporary file will be created in the default directory, unless there is a temporary directory specified with the environment variable PKTMP. For optimal performance, set this to point to a RAM Drive.

7) As PKZIP reads each file, it computes a CRC value for it. This CRC value is stored as part of the information concerning the file.

CRC

A CRC, or Cyclic Redundancy Check, is a value which represents a file. This value is arrived at by sending the data of the file through an algorithm. This algorithm starts with a base number, and then modifies it as the data passes through.

When the file is complete, the value of this number represents the CRC of the file. The CRC computed by PKZIP is a CRC of the uncompressed file, not the compressed data.

The resolution of the CRC check is very high. It can catch a single BIT difference between the original file and a modified file.

The .ZIP File Structure

8) When PKZIP adds the compressed file to the .ZIP file, it first writes out a "Local Header" about the file. This Header contains useful information about the file, including:

- The minimum version of PKZIP needed to extract this file
- The compression method used on this file
- File time
- File date
- The CRC value
- The size of the compressed data
- The uncompressed size of the file
- The filename

9) After PKZIP has written all of the files to disk, it appends the "Central Directory" to the end of the .ZIP file. This Directory contains the same information as the Local Header for every file, as well as additional information.

Some of this additional information includes:

- The version of PKZIP that created the file
- A comment about each file (if any)
- File attributes (Hidden, Read Only, System)
- The .ZIP file comment (if any)



A detailed specification for these structures is available in the file "appnote.txt" which is distributed with every copy of PKZIP.

Deleting Files From a .ZIP File

When PKZIP deletes files from a .ZIP file, it is done in this way:

- 1) PKZIP reads in the names of all the files contained in the .ZIP file.
- 2) PKZIP compares this list against the files you wish to delete.
- 3) Whatever files remain are moved into a new .ZIP file.
- 4) The original .ZIP file is deleted.

The original .ZIP file is not deleted until the new .ZIP file is created.

This means that in order to delete files from a .ZIP file, you must have enough disk space to hold the existing .ZIP file as well as the .ZIP file that has the deleted files removed.

Adding to an Existing ZIP File

Adding files to a .ZIP file is basically the same as creating a .ZIP file, but with one difference. Before PKZIP begins to add files, it first reads in the files that were in the existing .ZIP file. These old files and the new files are then both written out to a new .ZIP file. Once the new .ZIP file is complete, the old one is deleted.

This means that there must be enough free space for the old .ZIP file as well as the new .ZIP file to co-exist until the procedure is completed.



In the case of modifying an existing .ZIP file, the temporary file is created in the default directory. If you wish this temporary file to be created on another drive due to speed or space considerations, use the -b option.

Trouble-Shooting

This section offers some hints to aid you in locating and solving problems. Part of this section includes PKZIP and PKUNZIP command line and configuration file options that apply only to locating problems.

Problem	Solution/Reason	See Also
"Error in ZIP, use PKZIPFIX" - When compressing a file.	If you are just creating a ZIP file for the first time and receive this error, your command line is wrong.	Use Tutorial
"Warning:Bad Command or File Name"	PKZIP or PKUNZIP is not in your path, or you have mis-typed the command.	Use Tutorial
Error message: "SYS1041: The name specified is not recognized as an internal or external command, operable program or batch file."	You have attempted to run a program in an OS/2 shell. The program is not in your path, or you have mis-typed the command.	Use Tutorial
CRC failure	Get a new copy of the file.	F.A.Q. page 86
File does not compress	File is already compressed or is encrypted	F.A.Q. page 82
Files are still there after being Zipped	PKZIP copies files by default, to move them use the -m option.	PKZIP -m page 35
How do I make a ZIP file larger than a single diskette?	Use the -& option.	<i>PKZIP Backup Guide</i>
Disk Full error	Either the .ZIP file you are creating is too large to fit on the floppy, or you do not have enough room to update an existing .ZIP file. Create a multi-volume .ZIP file or use an alternate temporary drive.	<i>PKZIP Backup Guide</i> PKZIP -b page 30
Can't find PKSFX	There is no program called PKSFX. PKSFX refers to a "Self-extracting .ZIP file". That is a .ZIP file that has been made into an executable file with the program ZIP2EXE	PKSFX / ZIP2EXE page 57
I have lots of memory but I'm getting an "Out of memory" error?	Although PKZIP uses XMS/EMS memory, there are limitations that will occur due to your base (conventional) memory being limited. Try to clear up as much conventional memory as possible. If you are attempting to compress more than 8,000 files at once you will usually receive this error.	"Using List Files" page 63
Can't re-create stored subdirectories or return files to their original subdirectories.	Make certain you are using the -d option. If you are and still have this problem, subdirectories may not have been stored in the ZIP file.	F.A.Q. A-3

-3 Disable 32-bit instruction usage

This option is available for users with 80386/80486 type computers. It is used to turn OFF the added 32-bit instructions and register usage. When 32-bit instructions are available, PKZIP will utilize them to allow the program to function even faster. However, if you are running a different program that may conflict with the use of 32-bit instructions, you can disable them by using this command.

32-bit instructions will be disabled only for the current PKZIP operation. They will then be enabled immediately after the command operation is completed.

You can also disable 32-bit instructions all the time by using the PKNO386 environment variable. To set the environment variable, type PKNO386=ON. If the string PKNO386 is present in the environment, then PKZIP (and PKUNZIP) will not use 32-bit instructions or registers.

To enable 32-bit instructions after they have been disabled, type:

SET PKNO386= .

If you have persistent problems with another piece of software interfering with PKZIP, and you cannot upgrade or discard that software, you may turn off 32-bit instructions in the configuration file. The pkware.ini option line is:

386=DISABLE

Note that the configuration file is read only by PKZIP. PKUNZIP can only be set permanently by use of the environment variable.

You may override the configuration file or environment variable on PKZIP by using a -3+ option. This will cause PKZIP to use 32-bit instructions regardless of the environment variable being set, or an entry in the configuration file.

-^ Echo the command line

This option is used to display the PKZIP command that you entered on the console. It is useful when you are debugging an application or batch file which calls PKZIP. The file specifications that you entered will be echoed on the console along with the command output.

In this example, technical information about all of the *.doc files included in the DOCS.ZIP file will be displayed. The information will be sorted by date, and the command line will be redisplayed on the screen above the output.

```
E:\>pkzip docs.zip -vd -^ *.doc
```

-+ Disable EMS usage

-- Disable XMS/UMB usage

To use EMS/XMS/UMB features PKZIP and PKUNZIP require:

- XMS version 2.0 or greater
- EMS LIM 4.0 or greater

An XMS driver is required in order to have UMB and HMA support available.

PKZIP/PKUNZIP will use up to 256K of EMS memory. If you are in a situation where you need to preserve EMS memory for other use, such as when running a multi-node BBS system, you may wish to disable this feature.

A conflict with your XMS/EMS driver can be eliminated by disabling EMS usage. Problems relating to EMS usage would manifest themselves as:

- Unexplainable machine lock-up.
- Receiving "Warning:Bad Table" and "File fails CRC check" for every file in a .ZIP file being extracted.

EMS or XMS usage can be turned off in the configuration file with the pkware.ini option:

EMS=DISABLE

XMS=DISABLE

If EMS has been disabled in the configuration file, you may force usage back on with the -++ option.

If XMS has been disabled in the configuration file, you may force usage back on with the --+ option.

-~ Disable Network detection

When this option is specified, Network specific operations are disabled. If you feel you are experiencing problems due to a conflict with the Network specific operations use this option.

If the problem is persistent enough that you wish to disable Network operations all the time, you may do so with a configuration file command:

NET=DISABLE

If Network support has been disabled in the configuration file, you may force usage back on with the -~+ option.

-) **Disable DPMI Support**

PKZIP and PKUNZIP require 32-BIT DPMI 0.90 or greater.

DPMI support uses an extra 6.5K of conventional or UMB memory. If you are extremely tight on memory you may wish to disable DPMI support.

If a failure occurs in the DPMI Support it can be disabled with this option.

DPMI=DISABLE

If DPMI has been disabled in the configuration file, you may force usage back on with the -)+ option.

-(**Use "Slow" MemCopy**

Some computers suffer from a flaw that causes them to have errors when performing simple MemCopy functions. This problem is generally caused by the manufacturer's use of inferior grades or speeds of RAM chips for the external CPU memory cache on the motherboard.

This behavior will most commonly be seen on relatively fast machines (486/25 and higher).

This feature is only available on 486 (or higher) CPU's.

This problem will most frequently exhibit itself in the form of unexplainable CRC failures.

Use of this switch causes PKZIP to manipulate memory in a way that is less susceptible to this problem. Note that the added overhead in this process may slow down PKZIP by 20% to 40%.

SLOWMEMCOPY=ON

If normal MemCopy has been disabled in the configuration file, you may force usage back on with the -(+ option.

PKUNZIP has been designed to be mostly immune to this problem.

Please note that the external CPU cache has no relation to printer or disk caches.

PKWARE Customer Support Form

Please photocopy and complete this form as completely as possible before contacting our customer support. By providing this information you help us to provide assistance in an efficient and exact manner.

It is recommended that you first fax this form, then call PKWARE customer support. Indicate on this form that you will be calling.

If sending your problem by fax only, please specify how you would like to be contacted.

Please Print or Type, use "fine" fax mode if possible.

PKWARE Inc.
9025 N. Deerwood Dr.
Brown Deer, WI 53223

(414)354-8699
(414)354-8559 FAX
(414)354-8670 BBS

Problem Description

Be sure to include the following. Use as many additional sheets as necessary

- What you are trying to do
- What do you expect to happen
- What did happen and why it is a problem
- Copies of CONFIG.SYS, AUTOEXEC.BAT and any error messages

Customer Support Form

Date	
------	--

Customer Information

Name	
Company	
Voice Phone#	
FAX Phone#	
Address	
City, State ZIP Country	
Email ID (any)	

Configuration Information

PKZIP Version	
DOS Version/Vendor	
Computer Model/Make	
BIOS Version/Make	
Network Version/Vendor	
Drive Model/Vendor	

Problem Description

[illegible]

ERROR MESSAGES

The following error messages may appear when using PKZIP. When the word “WARNING” appears in the message, program execution will continue. Otherwise the program will abort to DOS.

Each warning message is prefaced with an error code. This error code consists of the letter 'E' or 'W' and a number. The letter corresponds to whether the error indicates a “Warning” or “Error” condition. The number corresponds to the DOS error level that would be returned.

Errors are listed in order of return code or warning level.

A brief explanation of each message follows.

PKZIP Error Messages

PKZIP: (E01) Can't create: XXXXX.

The named file could not be created. Either the target directory is full or the file already exists and is read-only or locked by another application. On a Network you may receive this message if you do not have file creation rights in the directory.

PKZIP: (E[2,3]) XXXXX.ZIP - error in ZIP, use PKZIPFIX.

The named .ZIP file has a corrupted file index. Use PKZIPFIX to attempt reconstruction of the .ZIP file. In some cases the damage to the .ZIP file may be very extensive. If the file is too badly damaged PKZIPFIX will not be able to recover the file.

NOTE: Many novice users receive this message because of improper command line syntax. If you specify a command line with only one file name following PKZIP and it is NOT a .ZIP file you will receive this message. PKZIP assumes that the file is a .ZIP file because you specified it as the first file name on the command line, and it attempts to place all files from the current directory into that file. Upon examination of the file it will not find any .ZIP file information (due to its not being a .ZIP file) and will then report this message. Please see the tutorial for further information.

PKZIP: (E[4-11]) Insufficient memory.

Insufficient memory is available to process the .ZIP file. Try making more memory available to PKZIP. If this does not rectify the problem then the .ZIP file might be corrupted and PKZIPFIX should be used to fix the .ZIP file. If you are creating a new .ZIP file and receive this

message, it is due to an attempt to ZIP a large number of files. Reduce the scope of your command and try again. If you are using a LIST file in your PKZIP command you may be receiving this error because the LIST file is too large. See "List Files" for more information.

PKZIP: (E12) No file(s) found.

No matching files were found to list using the View option. This means the filespec you specified on the command line has no match within the .ZIP file.

PKZIP: (E12) No files specified for deletion!

The -d (Delete) option was specified but no filenames were given to delete. This option does NOT default to *.* if no filenames are given. See the -d option on page 32 for further information.

PKZIP: (E12) Nothing to do!

No matching files, or files with the specified attributes, or files after the specified date etc., were found to compress. Often you will receive this message because the command line you entered is not correct. Check the exact name of the file you are specifying as well as the path, if any.

PKZIP: (E13) Can't find: XXXXX.ZIP.

The named .ZIP file could not be found. Either it does not exist or is not in the specified directory.

PKZIP: (E14) Insufficient disk space for ZIP comment.

There is insufficient disk space to hold the comment as entered. You will need to move the .ZIP file somewhere with more free space, or clear up more storage space in its current location. Note that a temporary copy of the .ZIP file is not created. The .ZIP file comment is appended directly to the existing .ZIP file.

PKZIP: (E14) Insufficient disk space for updated file: XXXXX.ZIP.

The -b option was used, and there is not enough space on the original drive containing the .ZIP file to receive the updated .ZIP file. (The update operation has added additional or larger files to the .ZIP file, causing it to grow larger than the original). Try to free up some space on the drive containing the .ZIP file, and retry the operation. Also, make sure that the drive specified with the -b option is different than the drive containing the .ZIP file.

PKZIP: (E14) Disk full, file: XXXXX.

A disk full error occurred while writing to the specified file. Try freeing up some disk space on the target drive. Also, see the -b option and the PKTMP environment variable.

PKZIP: (E15) Can't open XXXXX.ZIP for write access!

The named .ZIP file is read-only or locked by another application and can not be modified. This may also occur on a Network drive if you do not have adequate access to that file to allow you to modify it.

PKZIP: (E17) Too many files!

An attempt was made to compress more than 16,383 files at one time. This message could be received if you update a .ZIP file which has a corrupted central end directory.

PKZIP: (E24) Fatal EMS error: XXXXXXXXX

PKZIP: (E25) Fatal XMS error: XXXXXXXXX

An error has occurred in the interaction between PKZIP and your EMS or XMS handler. You may have a faulty EMS/XMS driver.

To be able to use PKZIP in spite of this problem, disable EMS/XMS usage with either the - + or - - option or the configuration file options EMS=DISABLE and XMS=DISABLE.

Please contact PKWARE Technical Support with a description of the situation in which the error occurred, the brand and version of your EMS driver, and the HEX code printed at the end of this error message.

PKZIP: (E26) Warning! DOS 3.0 or later needed to span disks

DOS versions below 3.0 do not support the features needed by PKZIP in order to span a .ZIP file across multiple disks. You will need to upgrade to a newer version of DOS in order to make use of this feature.

PKZIP: (E27) Warning! Non-removable or unsupported device

.ZIP files can only be spanned to a device which supports multiple storage units (i.e. multiple floppy disks can be inserted in series into a floppy drive). The device must also be a standard writable DOS device. .ZIP files may not be spanned onto a device such as a Hard drive which cannot be removed.

PKZIP: (E28) Destination for .ZIP file is same as temporary file

PKZIP: (E28) Source and destination is the same

PKZIP Warning Messages

PKZIP: (W01) Warning! Can't open XXXXX.CFG

When PKZIP runs it looks for a configuration file. The three places that PKZIP will look for its configuration file are: The current directory, the directory where PKZIP.EXE is located, and the directory pointed to by the environment variable PKZIP.CFG.

If the PKZIP.CFG environment variable is set, and PKZIP cannot locate a configuration file in any of the three places it looks, you will receive this error message.

PKZIP: (W02) Warning! Can't open XXXXX

This error usually occurs on Networks where you either do not have read access to a file or another process/user has the file open in an exclusive mode.

If the file is held open by another process you may be able to overcome this problem by using the `--` switch.

PKZIP will exit with an error level of 18 when this warning is received.

PKZIP: (W03) Warning! Can't delete XXXXX.

The `-m` (Move) option was specified to delete files after the .ZIP file was constructed. However, the named file could not be deleted, and is probably read-only. You may also receive this message if the file is on a network drive and you do not have access to delete the file, or it is locked by another application.

PKZIP: (W07) Warning! Bad System Track

PKZIP: (W08) Warning! Can't write FAT

PKZIP: (W09) Warning! Can't write root directory

PKZIP: (W10) Warning! Can't write boot sector

PKUNZIP Error Messages

PKUNZIP: (E[4-8]) Warning! Insufficient memory.

Insufficient memory is available to process the .ZIP file. Try making more memory available to PKUNZIP. If this does not rectify the problem, then the .ZIP file might be corrupted, and PKZIPFIX should be used to fix the .ZIP file. The more files present in a .ZIP file, the more memory it will take to process the file.

In some cases the .ZIP file may be too complex. Too many files are contained within the .ZIP file for PKZIP to handle. This may occur if the .ZIP file is created under an operating system other than DOS.

PKUNZIP: (E09) Can't open: XXXXX.ZIP.

The named .ZIP or list file(s) could not be found. It does not exist, was spelled incorrectly, or is not in the specified directory.

PKUNZIP: (E09) Can't find: XXXXX

PKUNZIP was unable to locate the specified .ZIP or @list file. The filespec specified was in error or a disk error has occurred.

PKUNZIP: (E11) No file(s) found.

No files were found to extract, test, update or list. The filespec specified on the command line does not match any files present in the .ZIP file, or all files found could not be extracted due to an incorrect password or incompatible compression type.

PKUNZIP: (E50) Disk full, file: XXXXX.

There is not enough free room on the destination drive/directory for the file being extracted. The program will abort and exit to DOS.

PKUNZIP Warning Messages

PKUNZIP: (W[01,02]) Warning! File has bad table

The Imploded file being tested or extracted has an error in its encoding. The file is probably corrupt or not a .ZIP file. Files that have been damaged in this way cannot be recovered.

PKUNZIP: (W03) Warning! File has bad table

The Reduced file being tested or extracted has an error in its encoding. The file is probably corrupt or not a .ZIP file. Files that have been damaged in this way cannot be recovered.

PKUNZIP: (W04) Warning! XXXXX.ZIP - error in ZIP, use PKZIPFIX.

The named .ZIP file has a corrupted file index. Use PKZIPFIX to reconstruct the .ZIP file. You may also receive this message if you attempt to extract a file which is not a .ZIP file.

If the file is less than 22 bytes in length it cannot be a .ZIP file and the "use PKZIPFIX" message will not be shown.

PKUNZIP: (W[05-07]) Warning! Insufficient memory

Insufficient memory is available to process the .ZIP file. Try making more memory available to PKUNZIP. If this does not rectify the problem, then the .ZIP file might be corrupted, and PKZIPFIX should be used to fix the .ZIP file. The more files present in a .ZIP file, the more memory it will take to process the file.

In some cases the .ZIP file may be too complex. Too many files are contained within the .ZIP file for PKZIP to handle. This may occur if the .ZIP file was created under an operating system other than DOS.

PKUNZIP: (W08) Warning! Can't open XXXXX.ZIP

PKUNZIP: (W09) Warning! Can't open printer device

The named file or printer device could not be opened. Either a disk error occurred, or the file is locked by another application. You may receive this error on a network drive if you do not have read access to that file.

The printer device may not exist, or may be off-line.

PKUNZIP: (W10) Warning! Can't create: XXXXX.

PKUNZIP: (W10) Warning! Invalid file name: XXXXX.

The named file could not be created. The output directory is invalid because it does not exist, is spelled incorrectly, or is full. You may also receive this message on a Network drive if you do not have file creation rights there or the directory is full.

In the case of an invalid file name error, the name stored in the .ZIP file is invalid. This may indicate a damaged .ZIP file, or one that has been tampered with.

PKUNZIP: (W11) Warning! I don't know how to handle: XXXXX.

The file XXXXX is compressed or encoded in a way that this version of PKUNZIP is unable to handle. Either a different version of PKUNZIP is required to properly extract this file, or the .ZIP file is possibly corrupted.

PKUNZIP: (W12) Skipping encrypted file.

Only the files that are password protected will be skipped (not extracted) because they are encrypted and the correct password was not entered.

PKUNZIP: (W13) Warning! Inconsistent local header for file: XXXXX.

The local header for file XXXXX is different than the central header information. Use PKZIPFIX to reconstruct the .ZIP file.

PKUNZIP: (W14) Incorrect password for file.

The .ZIP file is password protected, and cannot be opened unless the correct password is entered. The password entered is not the correct password. See "Passwords" in the Advanced Features section for further information.

PKUNZIP: (W15) Warning! File fails CRC check.

The CRC-32 check for the file being extracted or tested did not match the stored value for the file. The file is probably corrupted. See the "Trouble Shooting" section for more details.

PKUNZIP: (W16) Warning! File fails Authenticity Verification check

The .ZIP file is AV encoded but the AV could not be verified. Failure of AV indicates a file which has been tampered with or damaged. If the file has failed the AV check the contents are suspect.

PKUNZIP: (W17) Warning! File fails Authenticity Verification check

The .ZIP file is AV encoded and has been tampered with or damaged. File should be discarded and a new copy obtained.

PKUNZIP: (W18) Warning! File XXXXX already exists. Overwrite (y/n/a/r)?

The file XXXXX already exists on the disk. Pressing N will leave the original file on the disk, and not extract the file from the .ZIP file. Pressing Y will extract the file from the .ZIP file, and overwrite the file on the disk. If you want to overwrite this file, and ALL FOLLOWING FILES, press A for ALL. If you wish to extract this file, but not overwrite an existing file, you may press R for RENAME.

PKUNZIP will prompt you for a new name for the file. See Also: -o and -n options.

PKUNZIP: (W19) Warning! Invalid Shrunk file

The Shrunk file being tested or extracted has an error in its encoding. The file is probably corrupt or not a .ZIP file. Files that have been damaged in this way cannot be recovered.

PKUNZIP: (W20) Warning! Insufficient memory

Insufficient memory is available to process the .ZIP file. Try making more memory available to PKUNZIP. If this does not rectify the problem, then the .ZIP file might be corrupted, and PKZIPFIX should be used to fix the .ZIP file. The more files present in a .ZIP file, the more memory it will take to process the file.

PKUNZIP: (W21) Warning! File has bad table.

The Deflated file being tested or extracted has an error in its encoding. The file is probably corrupt or not a .ZIP file. Files that have been damaged in this way cannot be recovered.

PKUNZIP: (W23) Warning! File has bad table.

The Deflated file being tested or extracted has an error in its encoding (bad block type). The file is probably corrupt or not a .ZIP file. Files that have been damaged in this way cannot be recovered.

PKUNZIP: (W24) Warning!

You need PKUNZIP version: X.X or later to extract XXXXX

The file XXXXX is compressed or encoded in a way that this version of PKUNZIP is unable to handle. A more recent version of PKUNZIP is required to properly extract this file. The version number is listed as X.X. Contact PKWARE for upgrade information.

PKUNZIP: (W25) Warning! .ZIP file comment too big

The file comment attached to this .ZIP file is too large. The .ZIP file can still be handled normally but will not be displayed.

PKUNZIP: (W26) XXXXX.ZIP has errors.

The named .ZIP file had one or more errors detected in it. This message is displayed after PKUNZIP is finished acting upon a file. You may receive this message if you attempt to extract a file that is not a .ZIP file.

ZIP2EXE Warning/Error Messages

ZIP2EXE: (E01) Error in .ZIP file

ZIP2EXE: (E02) Error in .ZIP file, use PKZIPFIX

The .ZIP file you are attempting to process is corrupt. You may attempt recovery of the file by using PKZIPFIX. PKZIPFIX will work on either standard .ZIP files or self-extracting .EXE files.

ZIP2EXE: (E03-04) Insufficient memory to process file

The .ZIP file you are attempting to process is too complex. This may indicate a .ZIP file which contains too many files. In some cases you may receive this error because the file being processed is corrupt.

ZIP2EXE: (E07) Can't open Source file

The source .ZIP or .EXE file specified on the command line cannot be opened due to a disk error or insufficient network access.

ZIP2EXE: (E08) Error in .ZIP file

The .ZIP file you are attempting to process is corrupt. You may attempt recovery of the file by using PKZIPFIX. PKZIPFIX will work on either standard .ZIP files or self-extracting .EXE files.

ZIP2EXE: (E09) Disk full writing target file during EXE to ZIP process

ZIP2EXE: (E11) Disk full writing target file during ZIP to EXE process

During the process of converting a file there must be enough disk space available to contain both the original file and the file which will be created. If there is not enough room to process the file on the current drive you may direct the output to another drive by specifying it on the command line.

ZIP2EXE: (E12) Compression method not supported

The .ZIP file being converted contains a file compressed with a method not supported by the version of PKSFX being attached by ZIP2EXE.

You can only perform the ZIP2EXE process on .ZIP files created using the compression method(s) supported by the self-extracting module you are attaching.

ZIP2EXE: (W01) Warning! The file already exists, overwrite (y/n)?

The file already exists on the disk. Pressing N will not overwrite the original file. Pressing Y will overwrite the file.

PKSFX Warning/Error Messages

PKSFX: (W01) File fails CRC check

PKSFX: (W01) Unknown compression method

A warning may indicate any of several problems. See the corresponding PKUNZIP error message for explanation.

PKSFX: (E02) Can't find .EXE file

PKSFX: (E02,E03) Error in .EXE file

The self-extractor was unable to locate itself. When the self-extractor runs it must determine what the name of its executable file is. Anything which can prevent this from happening will generate this error.

This error usually indicates an incorrect installation of DOS on the target machine.

PKSFX: (E[4-8]) Insufficient memory

Insufficient memory is available to process the .ZIP file. Try making more memory available to the Self-extractor. If this does not rectify the problem, then the self-extracting file might be corrupted, and PKZIPFIX should be used to fix the .EXE file. The more files present in a self-extracting .ZIP file, the more memory it will take to process the file.

PKSFX: (E09) Unable to locate listfile

The @list file specified on the command line could not be found.

PKSFX: (E13) Error in .EXE file

The self-extracting file is corrupt. This may occur due to a disk error, a transmission error, or a software problem.

PKSFX: (E50) Disk full

The self-extractor has run out of space while extracting the files it contains. You may either free up more disk space or direct the self-extractor to place the files on a different drive.

PKSFX: Warnings numbers are the same as PKUNZIP [9-18, 20, 21, 23, 24]

PKSFX/PKUNZIP Junior Error Messages

PKSFXJR: Warning error

A warning may indicate any of several problems. See the corresponding PKUNZIP error message for explanation.

PKSFXJR: Can't find .EXE file

The self-extractor was unable to locate itself. When the self-extractor runs it must determine what the name of its executable file is. Anything which can prevent this from happening will generate this error.

This error usually indicates an incorrect installation of DOS on the target machine.

PKSFXJR: Error in .EXE file

The self-extracting file is corrupt. This may occur due to a disk error, a transmission error, or a software problem.

PKSFXJR: File has a bad table

The compressed data is corrupt. This can indicate a damaged self-extracting file or data that was originally corrupt when the self-extracting file was created.

PKSFXJR: Can't create file

The self-extractor was unable to create a file it intended to extract. This may occur because you either do not have sufficient rights to a directory, or because all the directory entries are full.

PKSFXJR: Disk full

The self-extractor has run out of space while extracting the files it contains. You may either free up more disk space or direct the self-extractor to place the files on a different drive.

PKSFXJR: Too many files

A junior self-extractor may only contain 512 or fewer files. This error indicates either that more than 512 files are within the self-extractor or that the file is corrupt.

Error Levels

In order to enhance the power of using PKZIP and PKUNZIP from within another program or within a batch file, they return error levels. These error levels are listed below.



Please note that if testing error levels in a batch file, DOS tests the error level not for equality, but for greater than or equal to.

PKZIP:

- 0 No error.
- 1 Bad file name or file specification.
- 2,3 Error in .ZIP file.
- 4-11 Insufficient Memory.
- 12 No files were found to add to the .ZIP file, or no files were specified for deletion.
- 13 File not found. The specified .ZIP file or list file was not found.
- 14 Disk full
- 15 .ZIP file is read-only and can not be modified.
- 16 Bad or illegal parameters specified. (help screen displayed)
- 17 Too many files.
- 18 Could not open file.
- 19-23,29 Insufficient Memory
- 26 DOS 3.2 or later needed to span disks
- 28 Destination for .ZIP is same as temporary file, or Source and destination is the same
- 255 User pressed control-c or control-break

PKUNZIP:

- 0 No error.
- 1 Warning error (such as failed CRC check).
- 2,3 Error in .ZIP file.
- 4-8 Insufficient Memory.
- 9 File not found. No .ZIP files found.
- 10 Bad or illegal parameters specified. (help screen displayed)
- 11 No files found to extract/view etc.
- 50 Disk Full.
- 51 Unexpected EOF in .ZIP file.
- 255 User pressed control-c or control-break

ZIP2EXE:

- 0 No error
- 1 Error in .ZIP file
- 2 Error in .ZIP file, use PKZIPFIX
- 3-4 Insufficient memory to process file
- 6 Bad or illegal parameters specified. (help screen displayed)
- 7 Can't open Source file
- 8 Error in .ZIP file
- 9 Disk full writing target file during EXE to ZIP process
- 11 Disk full writing target file during ZIP to EXE process
- 12 .ZIP file being converted contains a file compressed with a method not supported by the version of PKSFX being attached by ZIP2EXE
- 13 User answered "No" in response to an "overwrite(y/n)?" prompt

PKSFX:

- 1 Warning error such as:
File fails CRC check, Unknown compression method.
- 2,3 Can't find .EXE file
- 4-8 Insufficient memory
- 9 Unable to located listfile
- 10 Bad or illegal parameters specified. (help screen displayed)
- 12 User aborted at comment pausing
"Extract now (y/n)?" prompt
- 13 Error in .EXE file
- 50 Disk full

PKSFX/PKUNZIP Junior:

- 1 Warning error such as:
File fails CRC check, Unknown compression method.
- 2 Can't find .EXE file
- 3 Error in .EXE file
- 4 File has a bad table
- 5 Can't create file
- 6 Disk full
- 7 Too many files
- 8 User aborted at comment pausing
"Extract now (y/n)?" prompt

PK Safe ANSI

PKSFANSI.COM

PKSFANSI (PK Safe ANSI) is a Terminate and Stay Resident program that disables ANSI Keyboard Key Reassignments, thereby preventing “ANSI bombs” embedded in any text file (such as README files) or output by any program.

Normally, ANSI sequences that redefine the keyboard could be hidden inside ANY text file or program, and could be executed completely unnoticed until it is too late. PKSFANSI intercepts calls to the ANSI.SYS or other ANSI device drivers, and filters out any keyboard reassignments, while allowing other ANSI sequences through unaltered.

If a keyboard key reassignment is attempted, PKSFANSI will intercept the sequence and discard it. PKSFANSI also will BEEP to alert you that a reassignment was attempted.

PKSFANSI requires less than 1k bytes resident RAM, and should work with any ANSI driver, such as the standard ANSI.SYS driver, NANSI, ZANSI, DVANSI, etc. Note that if you use a memory resident ANSI driver, such as the DESQview DVANSI.COM driver, PKSFANSI should be loaded after the ANSI driver is loaded.

Usage

Add PK Safe ANSI to your AUTOEXEC.BAT command file. Early in your AUTOEXEC.BAT file add the line:

```
C:\PKWARE\PKSFANSI.COM
```

This will cause the PK Safe ANSI TSR to be loaded into memory, allowing you to safely use ANSI codes.

Once this is installed you may wish to turn on ANSI in your PKZIP configuration file. There should be this line present in your PKWARE.INI file:

```
ANSI=ENABLED
```

To turn on ANSI for a particular PKZIP operation, or for a PKUNZIP operation, use the command option -q.

```
pkunzip -v test.zip -q
```

Monochrome Option

PK Safe ANSI has one option. This option will cause PK Safe ANSI to discard color codes. This is useful if you are using a laptop computer with a monochrome display.

PKSFANSI -c

Please note that this only affects programs which use ANSI for their display control.

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